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(54) AUGMENTED REALITY DEVICE AND METHOD FOR DETECTING USER'S GAZE

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G02B 27/01 (2006.01)

(52) **U.S. Cl.**
CPC ***G06F 3/013*** (2013.01); ***G02B 27/0172*** (2013.01); *G02B 2027/0138* (2013.01); *G02B 2027/0178* (2013.01)

(58) **Field of Classification Search**
CPC ... G02B 27/01–0189; G02B 2027/0105–0198; G06F 3/01–018
See application file for complete search history.

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(57) **ABSTRACT**

An augmented reality (AR) device for detecting a user's gaze and a method thereof are provided. The AR device includes a waveguide; a light reflector including a pattern; a support configured to fix the AR device to the user's face of the AR device; a light emitter and a light receiver installed on the support; and at least one processor, wherein the at least one processor is configured to control the light emitter to emit light toward the light reflector, identify the pattern based on the light received through the light receiver, and obtain gaze information of a user of the AR device based on the identified pattern.

15 Claims, 21 Drawing Sheets

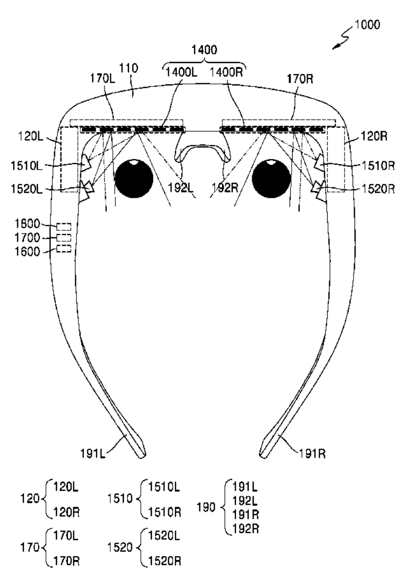

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FIG. 1

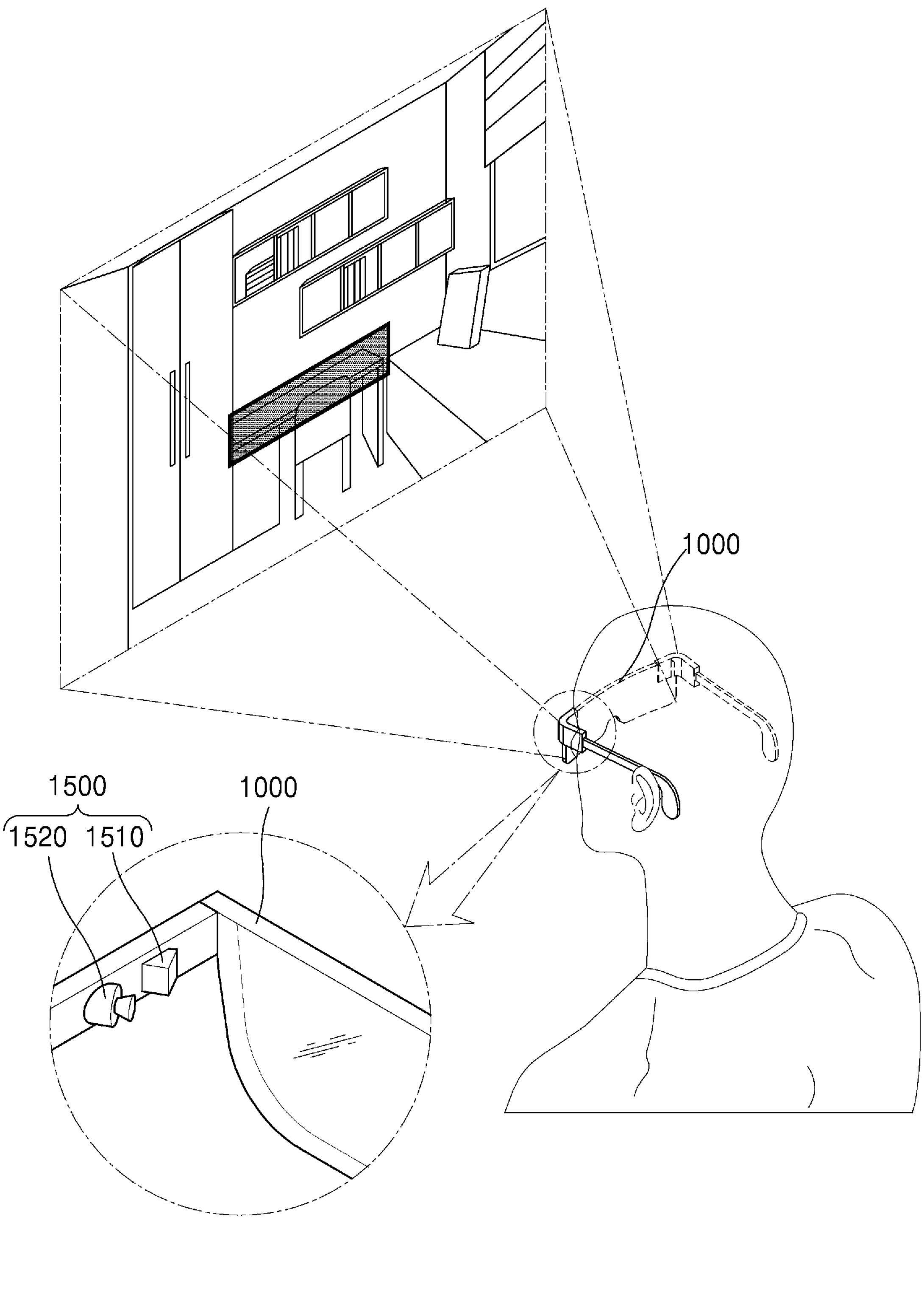
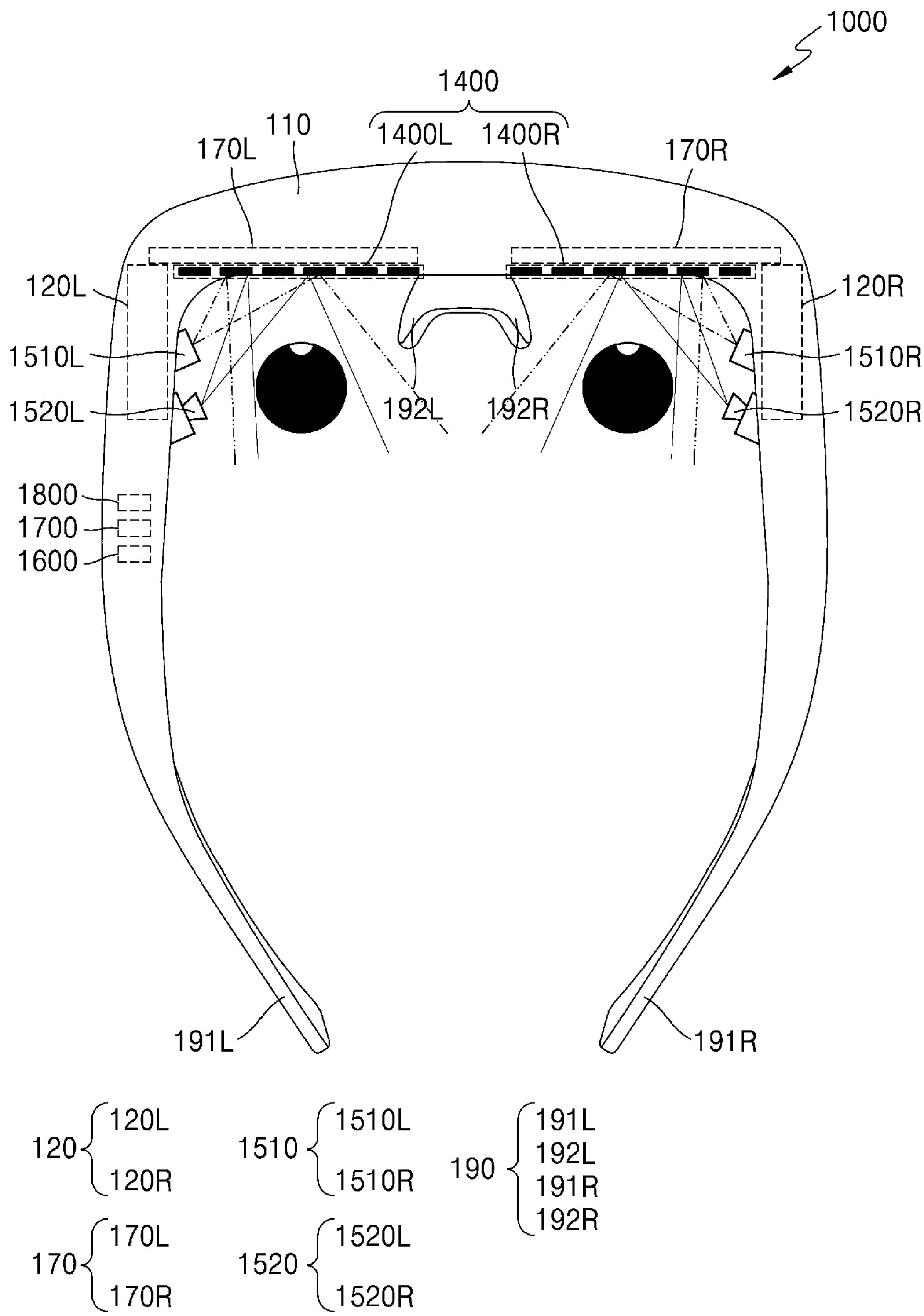


FIG. 2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1400
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170L
1400L
1400R
170R
120L
120R
1510L
1510R
1520L
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192R
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1800
1700
1600
191L
191R
120
120L
120R
1510
1510L
1510R
190
191L
192L
191R
192R
170
170L
170R
1520
1520L
1520R

FIG. 3

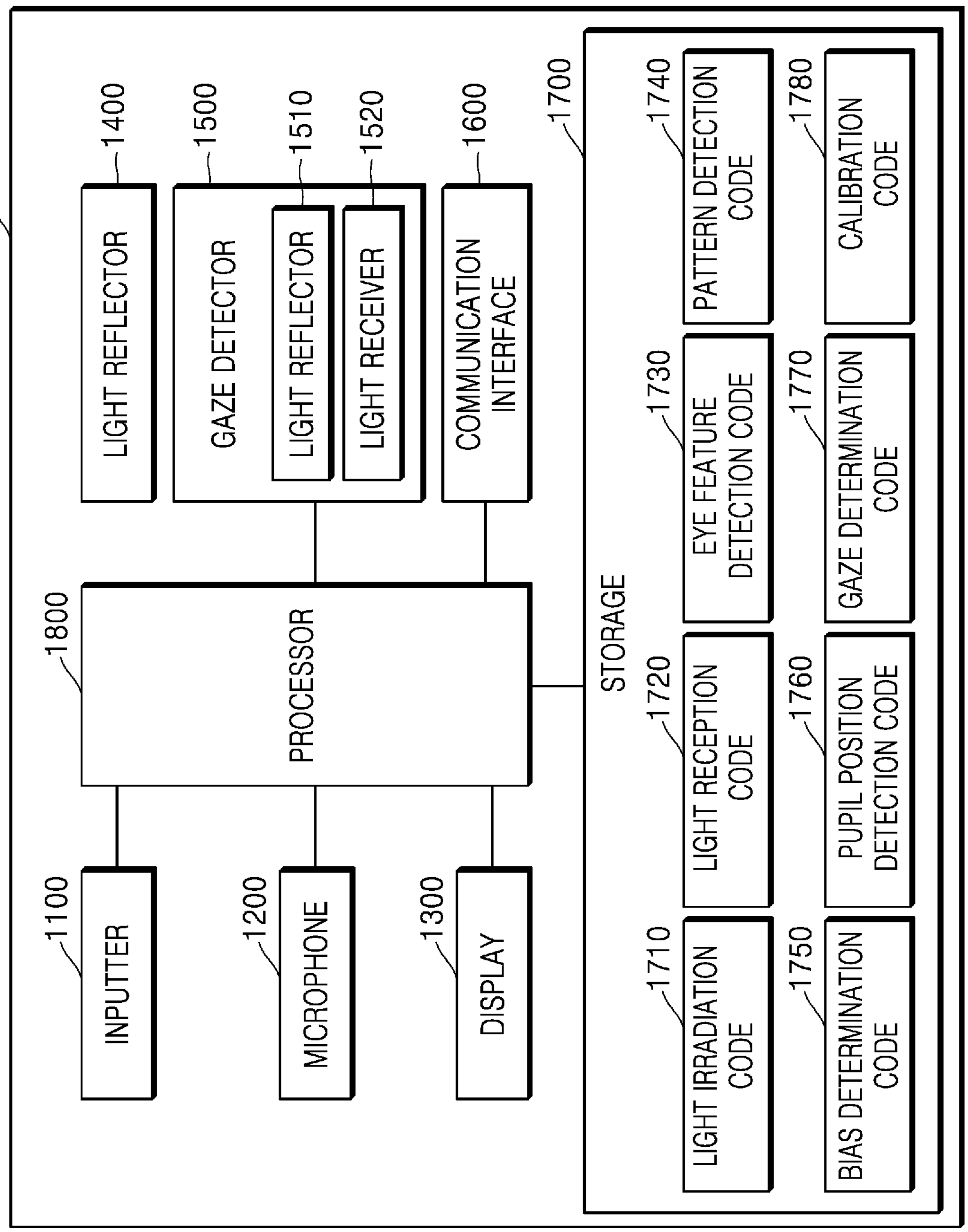

1000
INPUTTER
1100
MICROPHONE
1200
DISPLAY
1300
PROCESSOR
1800
LIGHT REFLECTOR
1400
GAZE DETECTOR
1500
LIGHT REFLECTOR
1510
LIGHT RECEIVER
1520
COMMUNICATION INTERFACE
1600
STORAGE
1700
LIGHT IRRADIATION CODE
1710
LIGHT RECEPTION CODE
1720
EYE FEATURE DETECTION CODE
1730
PATTERN DETECTION CODE
1740
BIAS DETERMINATION CODE
1750
PUPIL POSITION DETECTION CODE
1760
GAZE DETERMINATION CODE
1770
CALIBRATION CODE
1780

FIG. 4

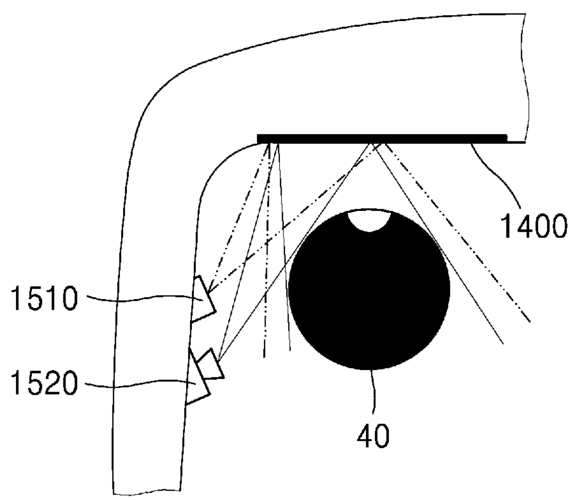


FIG. 5A

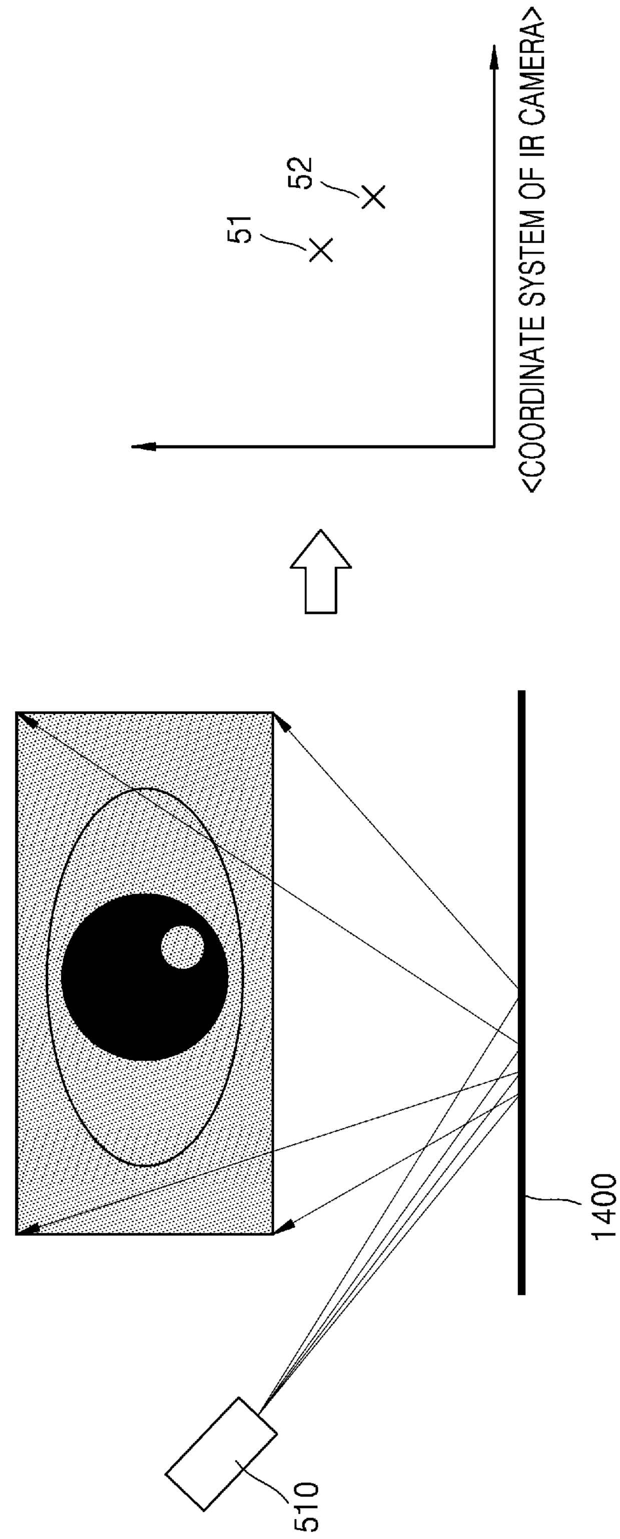


FIG. 5B

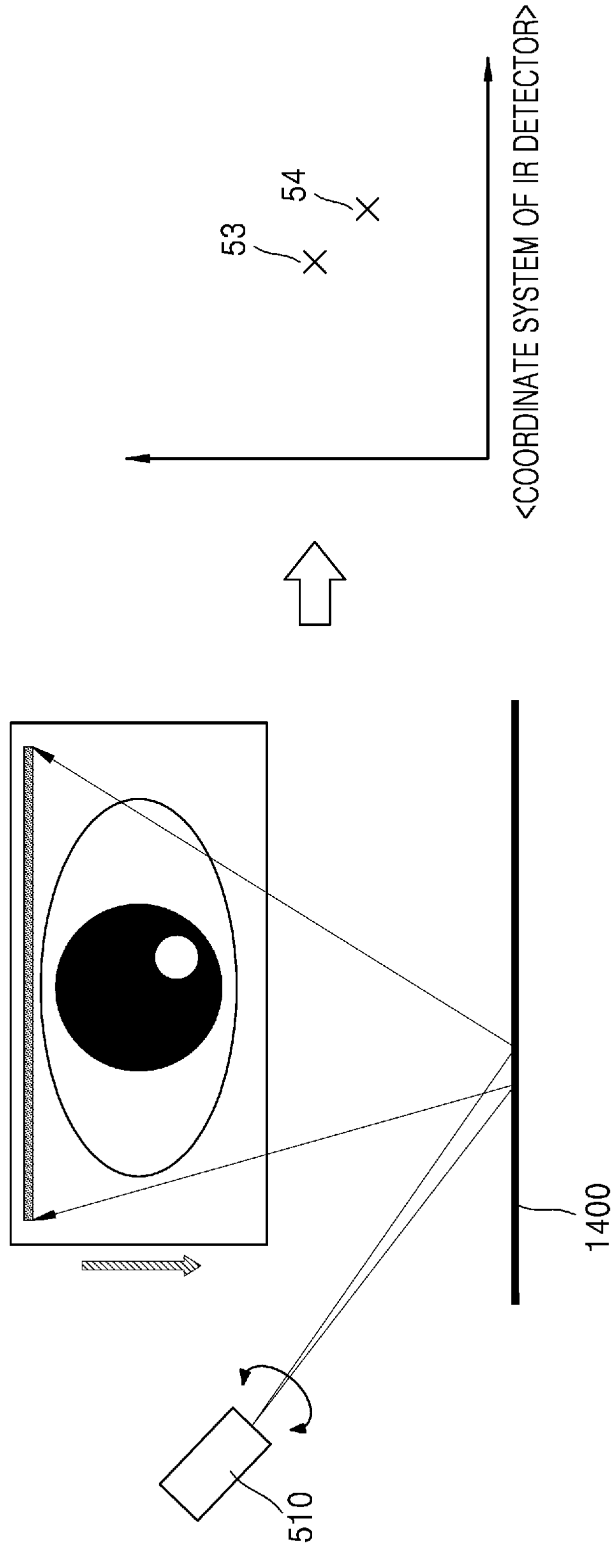


FIG. 5C

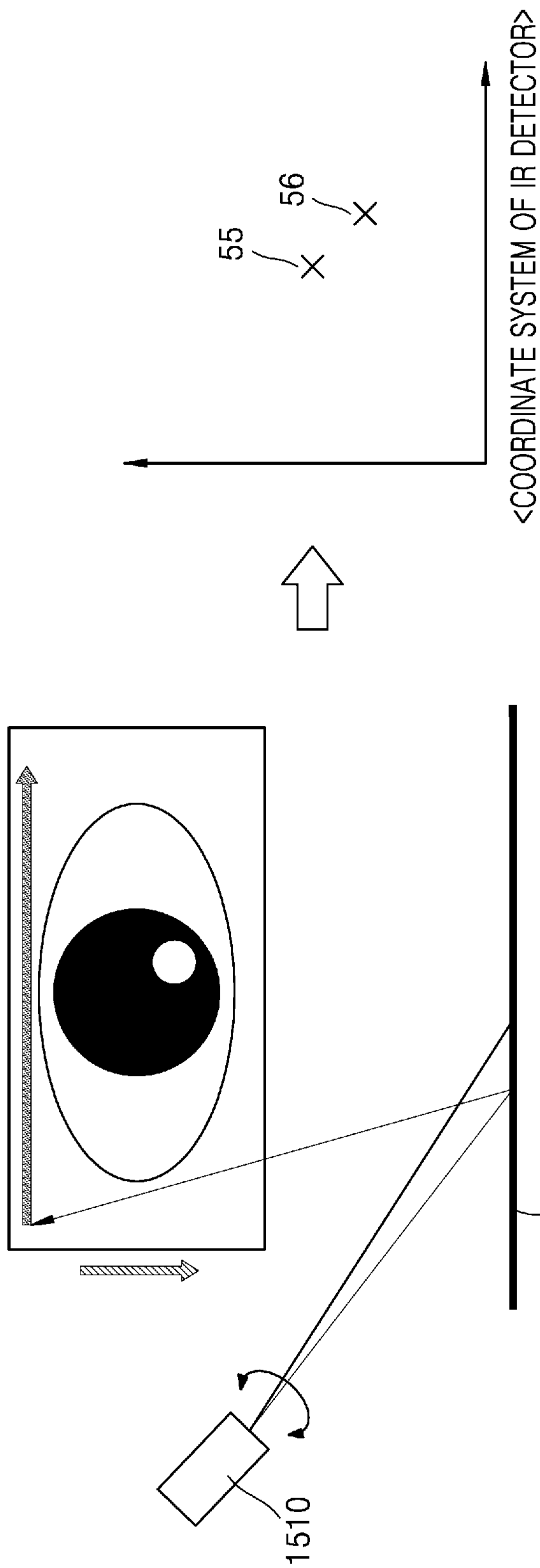


FIG. 6A

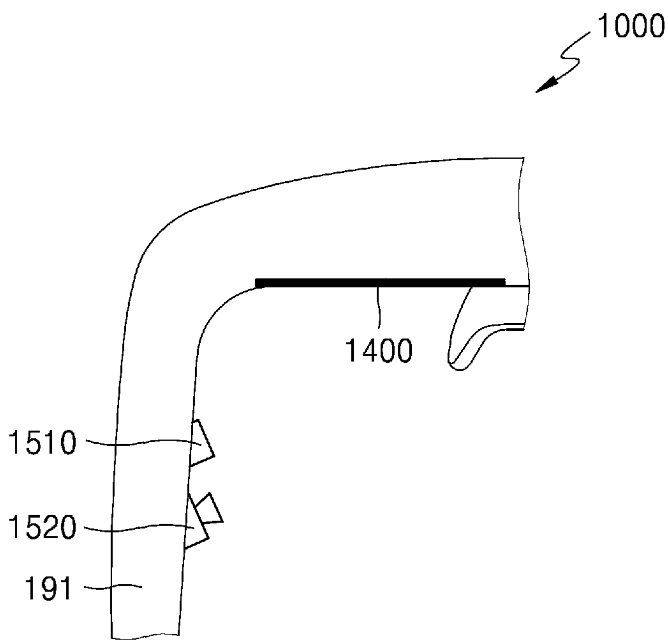


FIG. 6B

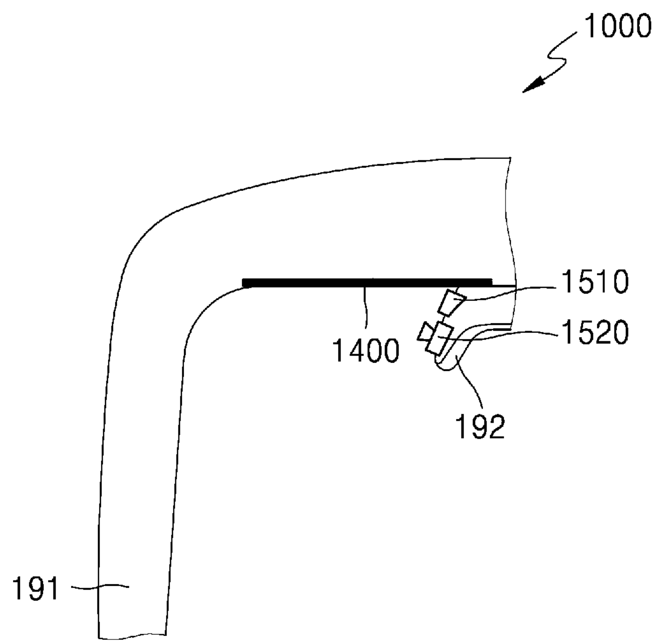


FIG. 6C

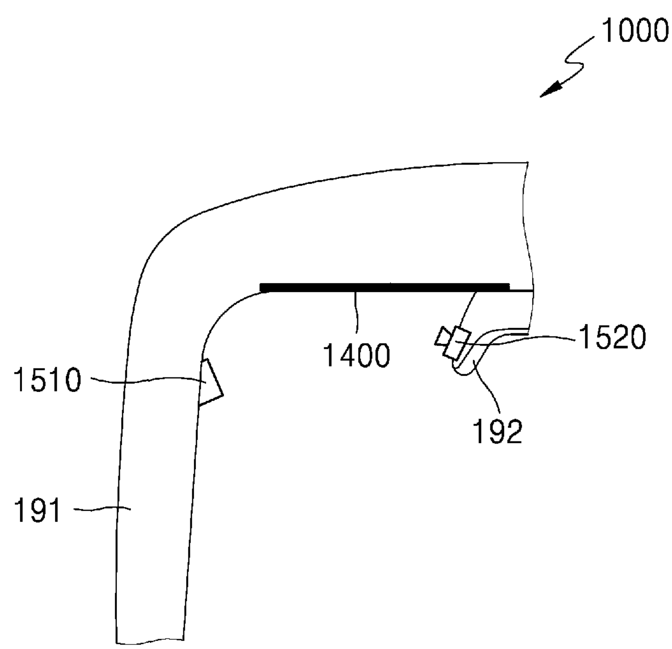


FIG. 6D

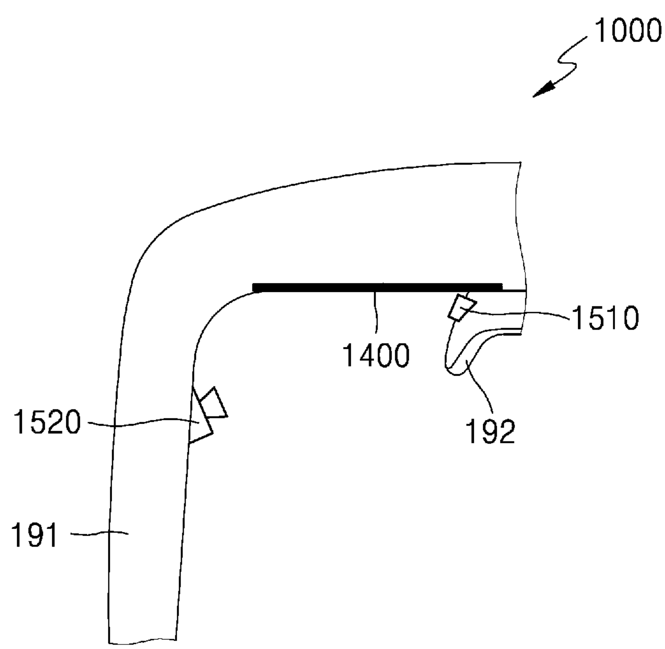


FIG. 7A

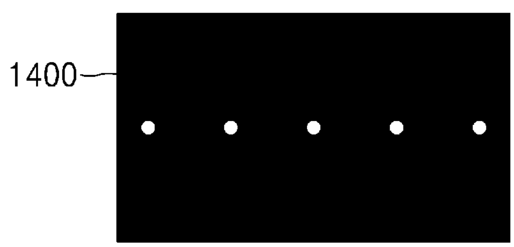


FIG. 7B

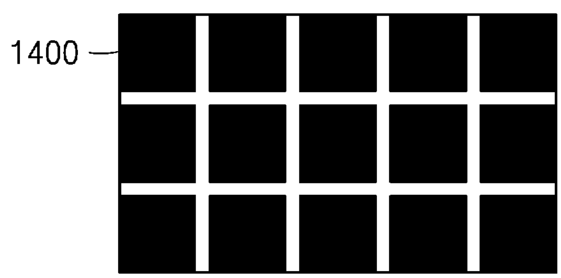


FIG. 7C

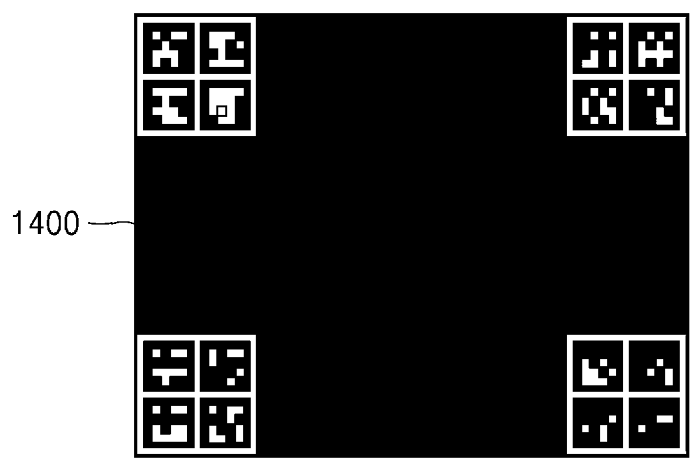


FIG. 7D

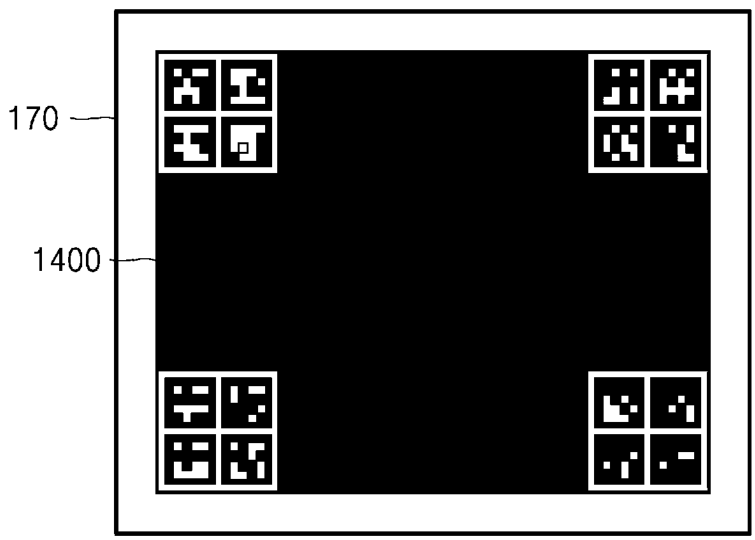


FIG. 8A

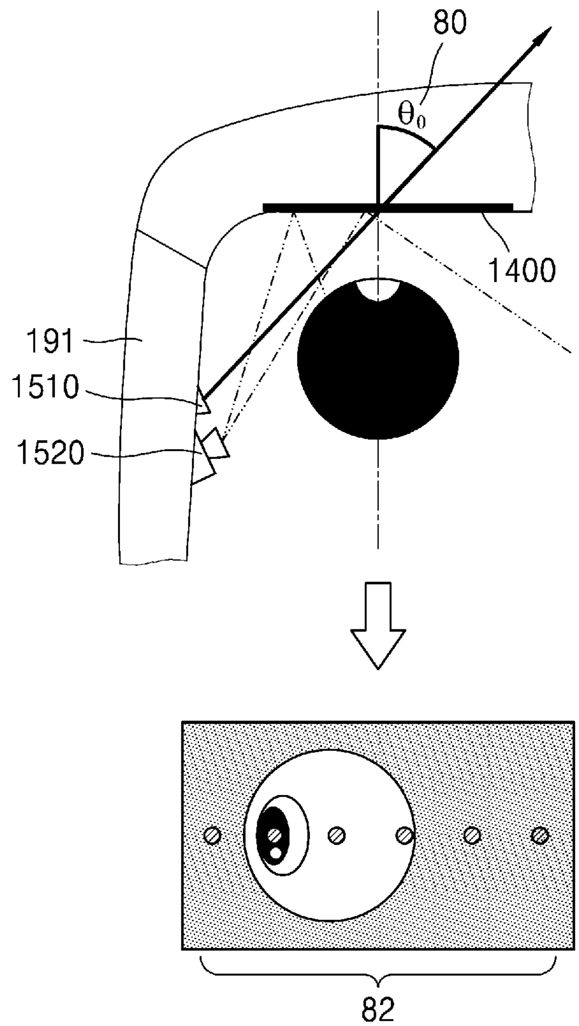


FIG. 8B

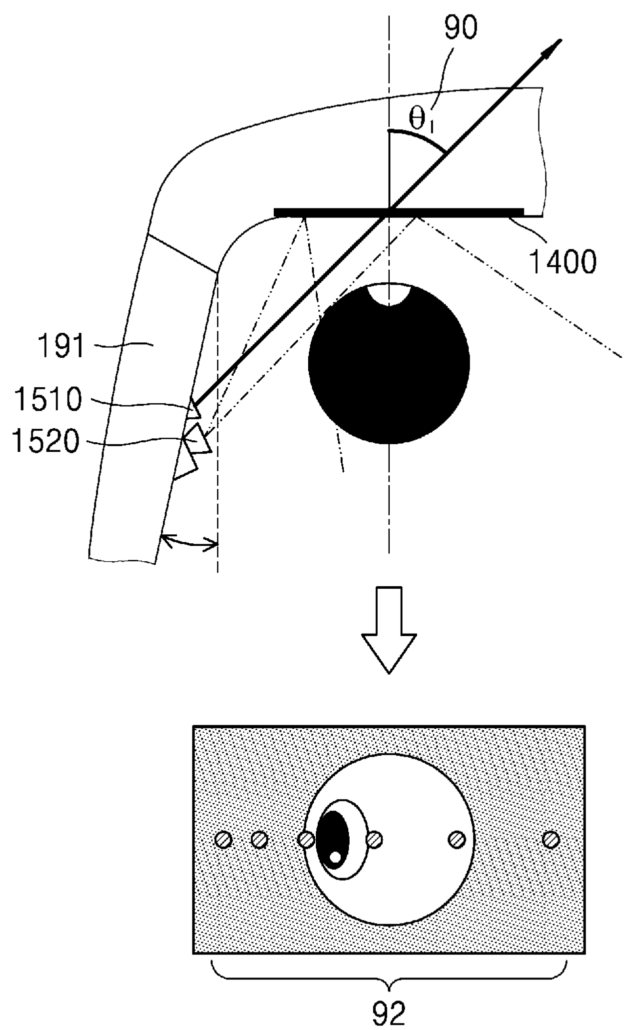
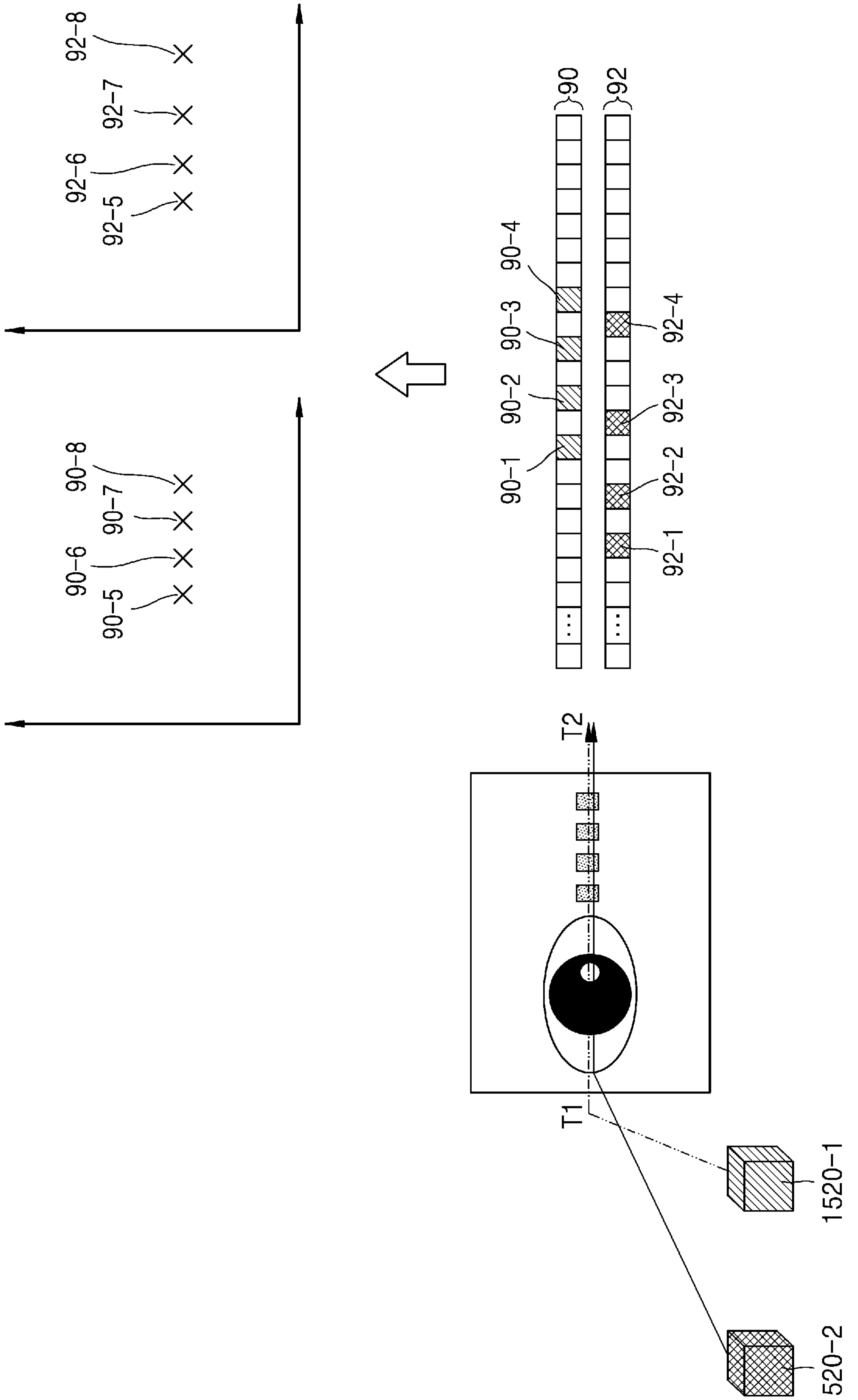


FIG. 9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92-6
92-7
92-8
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90
92
92-1
92-2
92-3
92-4
T1
T2
1520-1
1520-2

FIG. 10

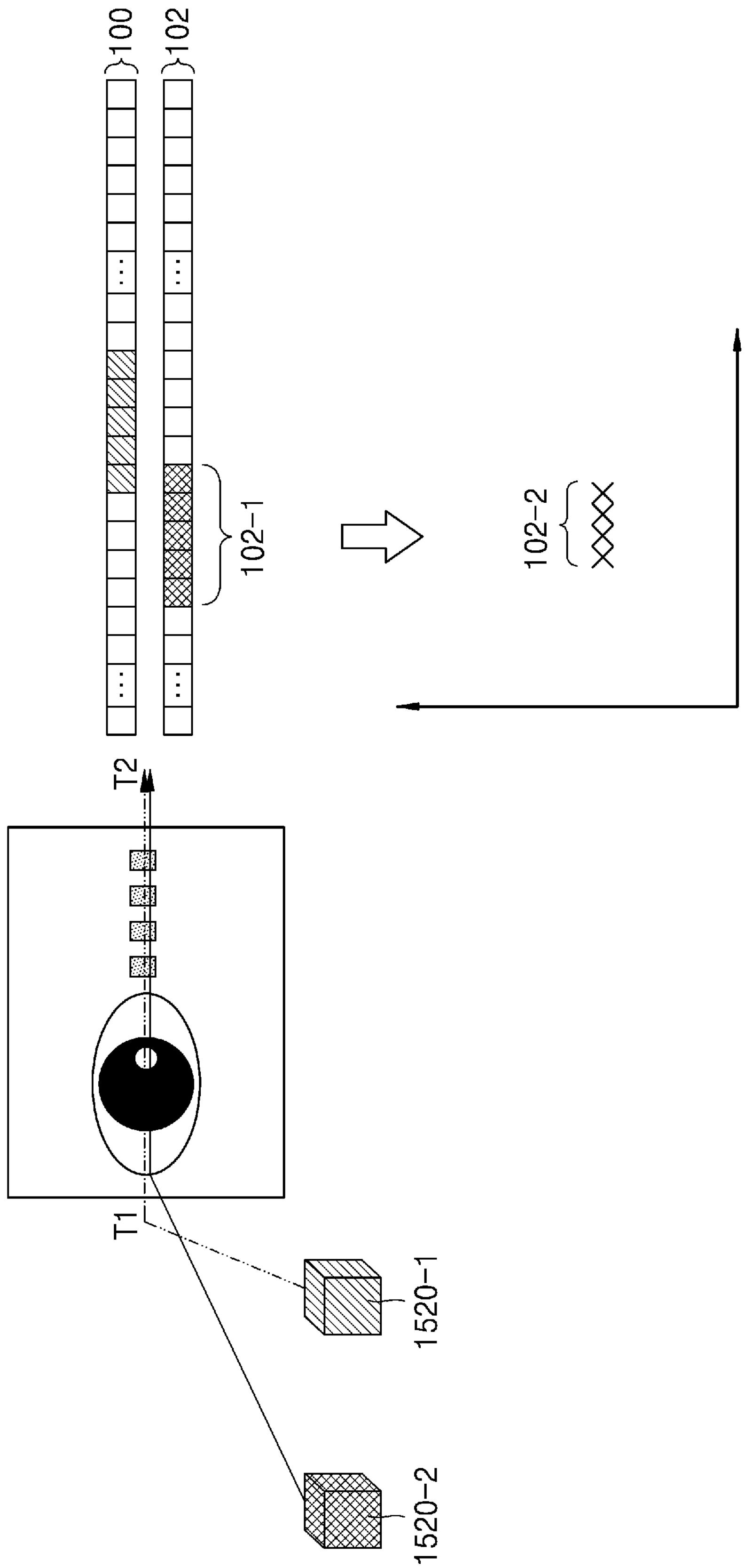


FIG. 11

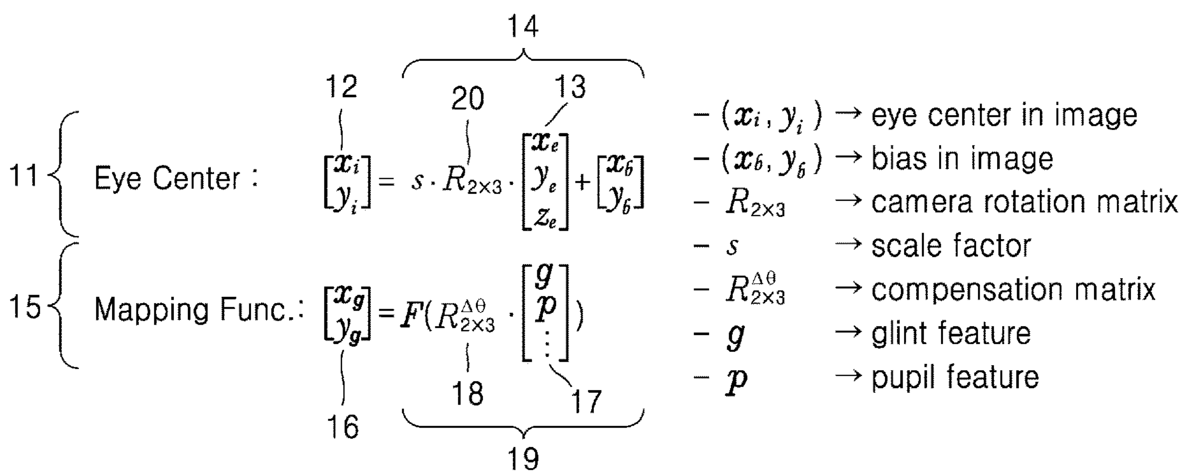
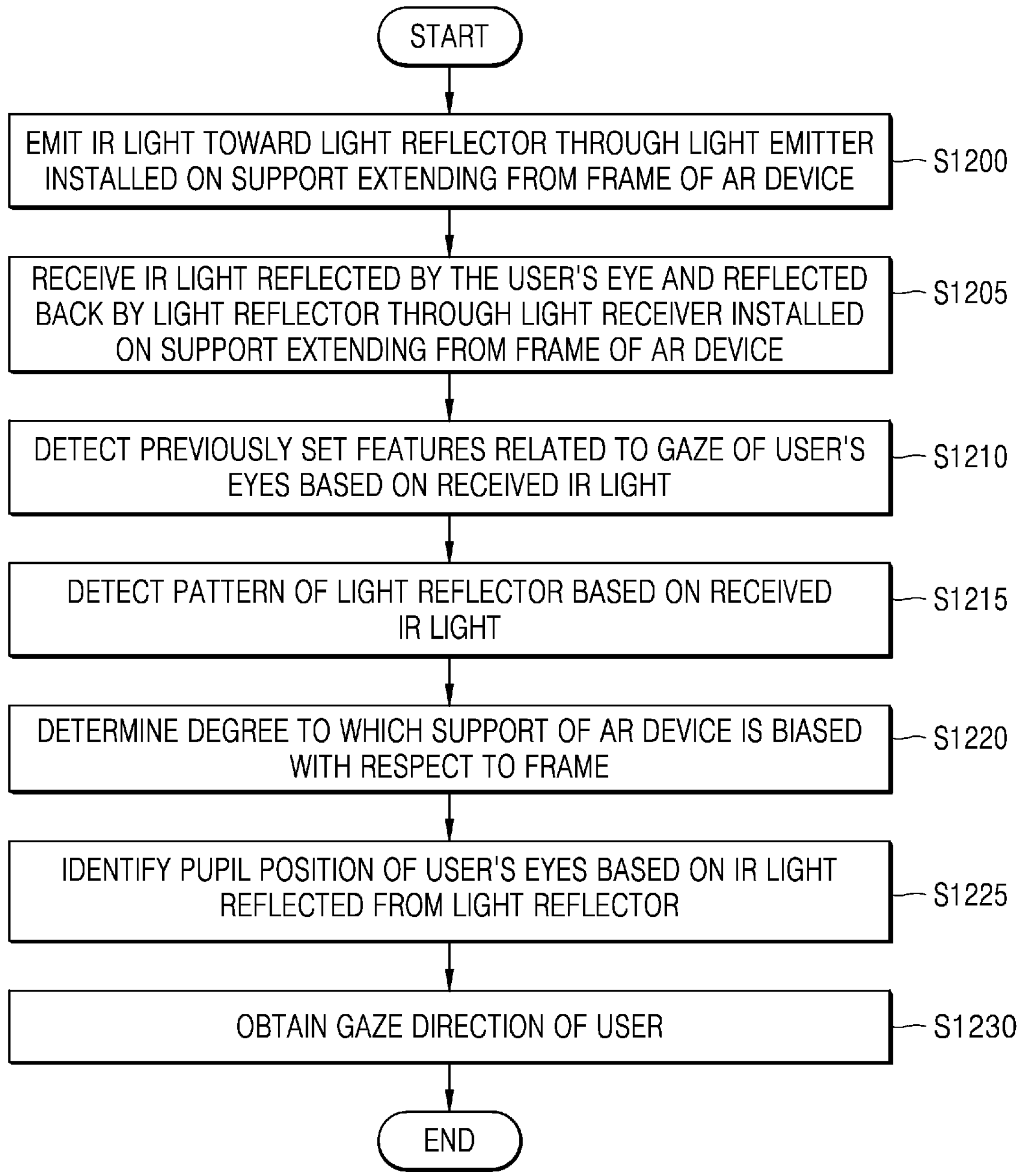

14
12
20
13
11
Eye Center :
15
Mapping Func.:
18
17
16
19
$\begin{bmatrix} x_i \\ y_i \end{bmatrix} = s \cdot R_{2\times3} \cdot \begin{bmatrix} x_e \\ y_e \\ z_e \end{bmatrix} + \begin{bmatrix} x_b \\ y_b \end{bmatrix}$
$\begin{bmatrix} x_g \\ y_g \end{bmatrix} = F(R^{\Delta\theta}_{2\times3} \cdot \begin{bmatrix} g \\ p \\ \vdots \end{bmatrix})$
- (x_i, y_i) → eye center in image
- (x_b, y_b) → bias in image
- $R_{2\times3}$ → camera rotation matrix
- s → scale factor
- $R^{\Delta\theta}_{2\times3}$ → compensation matrix
- g → glint feature
- p → pupil feature

FIG. 12


START
EMIT IR LIGHT TOWARD LIGHT REFLECTOR THROUGH LIGHT EMITTER INSTALLED ON SUPPORT EXTENDING FROM FRAME OF AR DEVICE
S1200
RECEIVE IR LIGHT REFLECTED BY THE USER'S EYE AND REFLECTED BACK BY LIGHT REFLECTOR THROUGH LIGHT RECEIVER INSTALLED ON SUPPORT EXTENDING FROM FRAME OF AR DEVICE
S1205
DETECT PREVIOUSLY SET FEATURES RELATED TO GAZE OF USER'S EYES BASED ON RECEIVED IR LIGHT
S1210
DETECT PATTERN OF LIGHT REFLECTOR BASED ON RECEIVED IR LIGHT
S1215
DETERMINE DEGREE TO WHICH SUPPORT OF AR DEVICE IS BIASED WITH RESPECT TO FRAME
S1220
IDENTIFY PUPIL POSITION OF USER'S EYES BASED ON IR LIGHT REFLECTED FROM LIGHT REFLECTOR
S1225
OBTAIN GAZE DIRECTION OF USER
S1230
END

AUGMENTED REALITY DEVICE AND METHOD FOR DETECTING USER'S GAZE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a bypass continuation of International Application No. PCT/KR2022/000682 designating the United States, filed on Jan. 13, 2022, in the Korean Intellectual Property Office and claiming priority to Korean Patent Application No. 10-2021-0008942, filed on Jan. 21, 2021, Korean Patent Application No. 10-2021-0084155, filed on Jun. 28, 2021, and Korean Patent Application No. 10-2021-0128345, filed on Sep. 28, 2021, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

TECHNICAL FIELD

The disclosure relates to an augmented reality (AR) device and method for detecting a user's gaze, and more particularly, to an AR device for detecting a user's gaze and a method thereof by using a light emitter and a light receiver located in a support of the AR device.

BACKGROUND ART

Augmented reality (AR) is a technology that projects a virtual image onto a physical environment space of the real world or a real world object and displays the virtual image as a single image. While being worn on a user's face or head, an AR device allows the user to see a real scene and a virtual image through a glasses type device using a see-through display such as a waveguide in front of the user's eyes. As research on such an AR device is being actively conducted, various types of wearable devices have been released or are expected to be released. In the glasses type AR device of the related art, a camera is generally arranged on a rim portion surrounding a waveguide to track the user's gaze, which causes the rim portion of the AR device to be enlarged, and further, causes the user wearing the AR device to feel uncomfortable.

DESCRIPTION OF EMBODIMENTS

Technical Problem

Provided are an augmented reality (AR) device and a method capable of detecting a user's gaze by using a light reflector and a light receiver located in a support extending from a frame of the AR device.

Provided are an AR device and a method capable of detecting a user's gaze by using light reflected through a light reflector formed on a waveguide.

Provided are an AR device and a method capable of more accurately detecting a user's gaze by calculating a degree of bias of a support of the AR device based on a pattern formed on a light reflector.

Solution to Problem

According to an aspect of the disclosure, there is provided an augmented reality (AR) device including: a waveguide; a light reflector comprising a pattern; a support configured to fix the AR device to a user's face of the AR device; a light emitter and a light receiver installed on the support; and at least one processor configured to: control the light emitter to emit light toward the light reflector, identify the pattern based on the light received by the light receiver, and obtain gaze information of a user of the AR device based on the identified pattern, wherein the light emitted toward the light reflector is reflected by the light reflector and directed toward an eye of the user, and wherein the light received by the light receiver comprises light from the light directed toward the eye of the user being reflected by the eye of the user.

The support may include a temple extending from a frame around the waveguide to be positioned on an ear of the user; and a nose support extending from the frame and positioned on a nose of the user.

The light reflector may be coated on the waveguide.

The light reflector may be formed on the waveguide.

The at least one processor may be further configured to analyze the identified pattern and identify a degree of bias of the support with respect to the frame, the support extending from the frame.

The at least one processor may be further configured to: generate a mapping function for calculating a position of a gaze point of the user based on the degree of bias of the support with respect to the frame, and based on the mapping function and the degree of bias of the support with respect to the frame, obtain the gaze information of the user.

The at least one processor may be further configured to, based on the light received by the light receiver, obtain a position of one or more feature points corresponding to the eye of the user.

The at least one processor may be configured to input the position of the one or more feature points corresponding to the eye of the user and the degree of bias of the support with respect to the frame into the mapping function and calculate the position of the gaze point of the user.

The position of the one or more feature points may include a position of a pupil feature point of the eye of the user and a position of a glint feature point of the eye of the user.

The at least one processor may be further configured to: display a target point at a specific position on the waveguide in order to calibrate the mapping function, receive light reflected by the eye of the user looking at the displayed target point through the light receiver, and calibrate the mapping function based on the light reflected by the eye of the user looking at the displayed target point.

The at least one processor may be further configured to: based on the light reflected by the eye of the user looking at the displayed target point, identify the pattern of the light reflector, based on the identified pattern, identify the degree of bias of the support, and based on the light reflected by the eye of the user looking at the displayed target point, obtain a position of one or more feature points corresponding to the eye of the user looking at the displayed target point.

The at least one processor may be further configured to input a degree of bias of the temple and the position of the one or more feature points into the mapping function, and calibrate the mapping function so that a position value of the target point is output from the mapping function.

The light emitter may be an infrared light-emitting diode (IR LED), and the light receiver is an IR camera.

The light emitter may be an infrared (IR) scanner, and the light receiver is an IR detector.

The at least one processor may be further configured to: based on IR light obtained from the IR detector, obtain a position of one or more feature points corresponding to the eye of the user calibrated according to a degree of bias of the support with respect to the frame, and obtain the gaze information of the user based on the calibrated position of the one or more feature points corresponding to the eye of the user.

According to another aspect of the disclosure, there is provided a method, performed by an augmented reality (AR) device, of detecting a user's gaze, the method including: emitting, by a light emitter installed in a support of the AR device, light toward a light reflector comprising a pattern, the light emitted by the light emitter being directed toward an eye of a user wearing the AR device, receiving, by a light receiver installed on the support, the light reflected by the eye of the user, identifying the pattern based on the light received through the light receiver, and obtaining gaze information of the user based on the identified pattern.

The method may further include analyzing the identified pattern and identifying a degree of bias of the support with respect to the frame based on the identified pattern, wherein the support extends from the frame, wherein the obtaining of the gaze information may include determining a gaze direction of the user based on the degree of bias of the support with respect to the frame.

The method may further include generating a mapping function for calculating a position of a gaze point of the user based on the degree of bias of the support with respect to the frame, wherein the obtaining of the gaze information may include calculating the position of the gaze point of the user based on the mapping function and the degree of bias of the support with respect to the frame.

The method may further include, based on the light received by the light receiver, obtaining a position of one or more feature points corresponding to the eye of the user, wherein the obtaining of the gaze information comprises inputting the position of the one or more feature points corresponding to the eye of the user and the degree of bias of the support with respect to the frame into the mapping function and calculating the position of the gaze point of the user.

According to another aspect of the disclosure, there is provided a computer-readable recording medium having recorded thereon a program for executing a method including: emitting, by a light emitter installed in a support of the AR device, light toward a light reflector comprising a pattern, the light emitted by the light emitter being directed toward an eye of a user wearing the AR device; receiving, by a light receiver installed on the support, the light reflected by the eye of the user; identifying the pattern based on the light received through the light receiver; and obtaining gaze information of the user based on the identified pattern.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a diagram illustrating an example in which an augmented reality (AR) device detects a user's gaze using a gaze detector located in a temple portion of the AR device, according to an example embodiment of the disclosure.

FIG. **2** is a diagram illustrating an example of an AR device according to an example embodiment of the disclosure.

FIG. **3** is a block diagram of an AR device according to an example embodiment of the disclosure.

FIG. **4** is a diagram illustrating an example of operations of a light emitter and a light receiver of an AR device, according to an example embodiment of the disclosure.

FIG. **5**A is a diagram illustrating an example of a light emitter that emits planar light, according to an example embodiment of the disclosure.

FIG. **5**B is a diagram illustrating an example of a light emitter that emits point light, according to an example embodiment of the disclosure.

FIG. **5**C is a diagram illustrating an example of a light emitter that emits line light, according to an example embodiment of the disclosure.

FIG. **6**A is a diagram illustrating an example in which a light emitter and a light receiver are arranged in a temple of an AR device, according to an example embodiment of the disclosure.

FIG. **6**B is a diagram illustrating an example in which a light emitter and a light receiver are arranged in a nose support of an AR device, according to an example embodiment of the disclosure.

FIG. **6**C is a diagram illustrating an example in which a light emitter and a light receiver are arranged in a temple and a nose support of an AR device, according to an example embodiment of the disclosure.

FIG. **6**D is a diagram illustrating an example in which a light emitter and a light receiver are arranged in a temple and a nose support of an AR device, according to an example embodiment of the disclosure.

FIG. **7**A is a diagram illustrating an example of a dot pattern formed on a light reflector of an AR device, according to an example embodiment of the disclosure.

FIG. **7**B is a diagram illustrating an example of a grid pattern formed on a light reflector of an AR device, according to an example embodiment of the disclosure.

FIG. **7**C is a diagram illustrating an example of a pattern in the form of a 2D marker, according to an example embodiment of the disclosure.

FIG. **7**D is a diagram illustrating an example of a light reflector that covers a part of a waveguide, according to an example embodiment of the disclosure.

FIG. **8**A is a diagram illustrating a light emission angle and a pattern before a temple of an AR device is biased, according to an example embodiment of the disclosure.

FIG. **8**B is a diagram illustrating a light emission angle and a pattern after a temple of an AR device, is biased according to an example embodiment of the disclosure.

FIG. **9** is a diagram illustrating an example of a pattern identified from an array of light received through a light receiver when a light emitter of an AR device is an infrared (IR) scanner, according to an example embodiment of the disclosure.

FIG. **10** is a diagram illustrating an example of an eye feature identified from an array of light received through a light receiver of the AR device when a light emitter of an AR device is an IR scanner, according to an example embodiment of the disclosure.

FIG. **11** is a diagram illustrating examples of functions used by an AR device to calculate a center of an eyeball and calculate a gaze point of a user, according to an example embodiment of the disclosure.

FIG. **12** is a flowchart of a method, performed by an AR device, of detecting a user's gaze, according to an example embodiment of the disclosure.

MODE OF DISCLOSURE

Throughout the disclosure, the expression "at least one of a, b or c" indicates only a, only b, only c, both a and b, both a and c, both b and c, all of a, b, and c, or variations thereof.

Hereinafter, embodiments of the disclosure will now be described in detail with reference to the accompanying drawings for one of skill in the art to be able to perform the disclosure without any difficulty. The disclosure may, however, be embodied in many different forms and should not be construed as being limited to the embodiments of the disclosure set forth herein. In order to clearly describe the disclosure, portions that are not relevant to the description of the disclosure are omitted, and similar reference numerals are assigned to similar elements throughout the present specification.

Throughout the specification, it will be understood that when an element is referred to as being "connected to" another element, it may be "directly connected to" the other element or be "electrically connected to" the other element through an intervening element. In addition, when an element is referred to as "including" a constituent element, other constituent elements may be further included not excluded unless there is any other particular mention on it.

The term 'augmented reality (AR)' herein denotes a technology that provides viewing of a virtual image on a physical environment space of the real world or viewing of a virtual image together with a real object.

In addition, the term 'AR device' denotes a device capable of creating 'AR', and includes not only AR glasses that are typically worn on a user's face but also includes head-mounted display (HMD) apparatuses and AR helmets that are worn on the user's head, etc.

Meanwhile, the term 'real scene' denotes a scene of the real world that the user sees through the AR device, and may include a real world object. In addition, the term 'virtual image' denotes an image generated by an optical engine, and may include both a static image and a dynamic image. The virtual image may be observed with a real scene, and may be an image representing information about a real object in the real scene, information about an operation of the AR device, a control menu, etc.

Accordingly, an AR device may be equipped with an optical engine to generate a virtual image including light generated by a light source, and a waveguide formed of a transparent material to guide the virtual image generated by the optical engine to the user's eyes and allow the user to see a scene of the real world together with the virtual image. In addition, as described above, the AR device needs to be able to allow the user to observe a scene of the real world, and thus, an optical element for redirecting the path of light that basically has straightness is required in order to guide the light generated by the optical engine to the user' eyes through the waveguide. Here, the path of the light may be redirected by using reflection by, for example, a mirror, or by using diffraction by a diffractive element, for example, a diffractive optical element (DOE) or a holographic optical element (HOE), but the disclosure is not limited thereto.

Hereinafter, the disclosure will be described in detail with reference to the accompanying drawings.

FIG. **1** is a diagram illustrating an example in which an augmented reality (AR) device **1000** detects a user's gaze using a gaze detector **1500** located in a temple portion of the AR device **1000** according to an example embodiment of the disclosure.

Referring to FIG. **1**, the AR device **1000** may detect the user's gaze by using a light emitter **1510** and a light receiver **1520**. The light emitter **1510** and the light receiver **1520** used to detect the user's gaze may be provided in, for example, the temple portion of the AR device **1000**, and the AR device **1000** may effectively identify user's eyes by using the light emitter **1510** and the light receiver **1520** provided in the temple portion. Infrared (IR) light may be emitted from the light emitter **1510** provided in the temple portion toward a waveguide of the AR device **1000**, reflected by a light reflector, and received from the user's eyes through the light receiver **1520** provided in the temple portion. Also, the AR device **1000** may obtain information about the user's eyes based on the received IR light, and detect a gaze direction of the user by using the obtained information about the eyes.

The AR device **1000** denotes a device capable of creating 'AR', and may include, for example, AR glasses that are worn on a user's face, but the disclosure is not limited thereto. For example, the AR device **1000** may include a head-mounted display (HMD) apparatus and an AR helmet that are worn on the user's head, etc. In this case, a gaze detector **1500** may be provided on an inner side part of the HMD apparatus facing the side of the user's eyes in the HMD apparatus or on an inner side part of the AR helmet facing the side of the user's eyes in the AR helmet.

FIG. **2** is a diagram illustrating an example of the AR device **1000** according to an example embodiment of the disclosure.

Referring to FIG. **2**, the AR device **1000** may include a glasses-type body configured to be worn by a user as a glasses-type display device.

The glasses-type body may include a frame **110** and a support **190**. The support **190** may extend from the frame **110** and be used to seat the AR device **1000** on a user's head. The support **190** may include a temple **191** and a nose support **192**. The temple **191** may extend from the frame **110** and may be used to fix the AR device **1000** to the user's head on a side surface of the glasses-type body. The nose support **192** may extend from the frame **110** and may be used to seat the AR device **1000** on a user's nose, and may include, for example, a nose bridge and a nose pad, but the disclosure is not limited thereto.

Also, a waveguide **170** to which a light reflector **1400** is attached may be located on the frame **110**. The frame **110** may be formed to surround an outer circumferential surface of the waveguide **170**. The waveguide **170** may be configured to receive projected light in an input region and output at least part of the input light in an output region. The waveguide **170** may include a left eye waveguide **170**L and a right eye waveguide **170**R.

A left eye light reflector **1400**L and the left eye waveguide **170**L may be provided at positions corresponding to a user's left eye, and a right eye light reflector **1400**R and the right eye waveguide **170**R may be provided at positions corresponding to a user's right eye. For example, the left eye light reflector **1400**L may be attached to the left eye waveguide **170**L, or the right eye light reflector **1400**R may be attached to the right eye waveguide **170**R, but the disclosure is not limited thereto. In addition, for example, the left eye light reflector **1400**L may be coated on the inner side of the left eye waveguide **170**L to be attached to the left eye waveguide **170**L, or the right eye light reflector **1400**R may be coated on the inner side of the right eye waveguide **170**R to be attached to the right eye waveguide **170**R.

In addition, an optical engine **120** of a projector that projects display light including an image may include a left eye optical engine **120**L and a right eye optical engine **120**R. The eye optical engine **120**L and the right eye optical engine **120**R may be located on both sides of the AR device **1000**. Alternatively, one optical engine **120** may be included in a central portion around the nose support **192** of the AR device **1000**. Light emitted from the optical engine **120** may be displayed through the waveguide **170**.

The light emitter **1510** and the light receiver **1520** of the gaze detector **1500** may be provided on an inner side part of the support **190** of the AR device **1000**, which is a position between the support **190** and user's eyes. The light emitter **1510** and the light receiver **1520** may be provided to face the light reflector **1400** in the support **190** of the AR device **1000**. For example, the light emitter **1510** and the light receiver **1520** may be provided at positions spaced from the frame **110** by about 10 mm to 15 mm on the inner side of the temple **191** of the AR device **1000**, in order to respectively emit and receive IR light without being disturbed by user's hair, etc.

FIG. **3** is a block diagram of the AR device **1000** according to an example embodiment of the disclosure.

Referring to FIG. **3**, the AR device **1000** according to an example embodiment of the disclosure may include a user inputter **1100**, a microphone **1200**, a display **1300**, a light reflector **1400**, a gaze detector **1500**, a communication interface **1600**, a storage **1700**, and a processor **1800**. Also, the gaze detector **1500** may include the light emitter **1510** and the light receiver **1520**.

The user inputter **1100** refers to a means by which a user inputs data for controlling the AR device **1000**. For example, the user inputter **1100** may include a key pad, a dome switch, a touch pad (e.g., a touch-type capacitive touch pad, a pressure-type resistive overlay touch pad, an infrared sensor-type touch pad, a surface acoustic wave conduction touch pad, an integration-type tension measurement touch pad, a piezoelectric effect-type touch pad), a jog wheel, a jog switch, but the disclosure is not limited thereto.

The microphone **1200** may receive an external audio signal, and process the received audio signal into electrical voice data. For example, the microphone **1200** may receive an audio signal from an external device or a speaker. The microphone **1200** may use various denoising algorithms for removing noise generated during a process of receiving the external audio signal. The microphone **1200** may receive a voice input of the user for controlling the AR device **1000**.

The display **1300** may display information processed by the AR device **1000**. For example, the display **1300** may display a user interface for capturing an image of surroundings of the AR device **1000**, and information related to a service provided based on the captured image of the surroundings of the AR device **1000**.

According to an example embodiment of the disclosure, the display **1300** may provide an AR image. The display **1300** according to an example embodiment of the disclosure may include the waveguide **170** and the optical engine **120**. The waveguide **170** may include a transparent material through which a partial region of a rear surface is visible when the user wears the AR device **1000**. The waveguide **1320** may be configured as a flat plate of a single layer or multi-layer structure including a transparent material through which light may be internally reflected and propagated. The waveguide **1320** may face an exit surface of the optical engine **120** to receive light of a virtual image projected from the optical engine **120**. Here, the transparent material is a material through which light is capable of passing, its transparency may not be 100%, and may have a certain color. According to an example embodiment of the disclosure, the waveguide **170** includes the transparent material, and thus the user may view not only a virtual object of the virtual image but also an external real scene, so that the waveguide **170** may be referred to as a see-through display. The display **1300** may output the virtual object of the virtual image through the waveguide **170**, thereby providing an AR image. When the AR device **1000** is a glasses type device, the display **1300** may include a left display and a right display.

The light reflector **1400** may reflect light emitted from the light emitter **1510** which will be described later. The light reflector **1400** and the waveguide **170** may be provided at positions facing the user's eyes, and may be attached to each other. For example, the light reflector **1400** may be coated on at least a partial region of the waveguide **170**. In addition, the light reflector **1400** may be attached to or coated on other elements included in the glasses type AR device **1000** in addition to the waveguide **170**, for example, a vision correcting lens for vision correction or a cover glass installed to protect the waveguide **170**. The light reflector **1400** may include a material capable of reflecting IR light emitted from the light emitter **1510**. The light reflector **1400** may be, for example, silver, gold, copper, ora material including one or more of these metals, but the disclosure is not limited thereto. Accordingly, the IR light emitted from the light emitter **1510** may be reflected by the light reflector **1400** and directed toward the user's eyes, and the IR light reflected back from the user's eyes may be reflected by the light reflector **1400** and directed toward the light receiver **1520**.

The light reflector **1400** may be coated on the waveguide **170** to have a certain pattern. The pattern formed on the light reflector **1400** may include, for example, a dot pattern, a line pattern, a grid pattern, a 2D marker, etc., but the disclosure is not limited thereto. In addition, the pattern formed on the light reflector **1400** may be formed on, for example, a part of the waveguide **170** at which the user's gaze is less frequently directed. The pattern formed on the light reflector **1400** may be formed on, for example, a part of the waveguide **170** that does not interfere with capturing or scanning the user's eyes. For example, the certain pattern may indicate a pattern formed by a part in which light emitted from the light emitter **1510** is reflected and a part in which the light is not reflected, in the light reflector **1400**. Because light emitted toward the part in which the light is not reflected, in the light reflector **1400** is not reflected by the light reflector **1400**, the light receiver **1520** does not receive the light emitted toward the part in which the light is not reflected. Accordingly, the pattern of the light reflector **1400** formed by the part in which light is reflected and the part in which the light is not reflected may be detected from the light received by the light receiver **1520**. In addition, when the light emitted from the light emitter **1510** is IR light, the certain pattern may include a material for reflecting the IR light, and the material for reflecting the IR light may not be visible to the user's eyes. Because most of a real world light or real scene observed by the user through the AR device **1000** includes visible light, the user may not be disturbed by the light reflector **1400** in which the certain pattern is formed, and may observe the real world light or real scene.

The gaze detector **1500** may include the light emitter **1510** that emits IR light for detecting the user's gaze and the light receiver **1520** that receives the IR light, and may detect data related to the user's gaze of the user who wears the AR device **1000**.

The light emitter **1510** of the gaze detector **1500** may emit the IR light toward the light reflector **1400** so that the IR light reflected by the light reflector **1400** may be directed toward the user's eyes. The light emitter **1510** may emit the IR light toward the light reflector **1400**, the emitted IR light may be reflected by the light reflector **1400**, and the reflected IR light may be directed toward the user's eyes. The light emitter **1510** may be provided at a position in the AR device **1000** where the IR light may be emitted toward the light reflector **1400**. The light emitter **1510** may be located on the support **190** of FIG. **2** that supports the AR device **1000** on the user's face, for example, like the temple **191** and the nose support **192** of FIG. **2**.

In addition, the IR light reflected from the user's eyes may be reflected by the light reflector **1400** and received by the light receiver **1520** of the gaze detector **1500**. The IR light directed toward the user's eyes may be reflected from the user's eyes, the IR light reflected from the user's eyes may be reflected by the light reflector **1400**, and the light receiver **1520** may receive the IR light reflected by the light reflector **1400**. The light receiver **1520** may be provided at a position in the AR device **1000** where the IR light reflected from the light reflector **1400** may be received. The light receiver **1520** may be located on the support **190** of FIG. **2** that supports the AR device **1000** on the user's face, for example, like the temple **191** and the nose support **192** of FIG. **2**. Also, for example, the nose support **192** of FIG. **2** may include a nose bridge and a nose pad. In addition, the nose bridge and the nose pad may be integrally configured, but the disclosure is not limited thereto.

For example, the light emitter **1510** may be an IR light-emitting diode (LED) that emits IR light, and the light receiver **1520** may be an IR camera that captures IR light. In this case, the IR camera may capture the user's eyes using the IR light reflected by the light reflector **1400**. When the light emitter **1510** is the IR LED and the light receiver **1520** is the IR camera, the light emitter **1510** may emit IR light of planar light toward the light reflector **1400**, and the light receiver **1520** may receive the IR light of the planar light reflected from the light reflector **1400**. The planar light may be light emitted in a planar form, and the planar light emitted from the light emitter **1510** may be directed toward at least a part of the entire region of the light reflector **1400**. At least a part of the entire region of the light reflector **1400** may be set so that the planar light reflected from at least a part of the entire region of the light reflector **1400** may cover the user's eyes.

Alternatively, for example, the light emitter **1510** may be an IR scanner that emits IR light, and the light receiver **1520** may be an IR detector that detects IR light. In this case, the IR scanner may emit IR light so that the IR light for scanning the user's eyes is directed toward the user's eyes, and the IR detector may detect the IR light reflected from the user's eyes. When the light emitter **1510** is the IR scanner that emits IR light and the light receiver **1520** is the IR detector that detects IR light, the light emitter **1510** may emit line lights in the form of line, and the line lights emitted from the light emitter **1510** may be directed toward a part of the entire region of the light reflector **1400**. At least a part of the entire region of the light reflector **1400** may be set so that the line lights reflected from at least a part of the entire region of the light reflector **1400** may cover the user's eyes. When the light emitter **1510** is the IR scanner that emits IR light and the light receiver **1520** is the IR detector that detects IR light, the light emitter **1510** may emit point lights in the form of point, and the point lights emitted from the light emitter **1510** may be directed toward a part of the entire region of the light reflector **1400**. At least a part of the entire region of the light reflector **1400** may be set so that the point lights reflected from at least a part of the entire region of the light reflector **1400** may cover the user's eyes.

When the light emitter **1510** is the IR scanner and the light receiver **1520** is the IR detector, the light emitter **151** may emit IR light of the point light or the line light to the light reflector **1400**, and the light receiver **1520** may receive the IR light of the point light or the linear light reflected from the light reflector **1400**. In this case, the light emitter **1510** may sequentially emit IR light while moving a light emitting direction of the light emitter **1510** so that the IR light of the point light or the line light may cover a space where the user's eyes are located. Although the IR scanner generally includes the IR LED and a micro-electro mechanical systems (MEMS) mirror capable of controlling a direction of the IR light emitted from the IR LED and reflecting the IR light, hereinafter, the IR scanner, the IR LED and the MEMS mirror are collectively referred to as and described as an IR scanner. In addition, although the IR detector generally includes several photodiodes installed in a part where light detection is required, hereinafter, the IR detector and photodiodes are described as the IR detector.

When the AR device **1000** is a glasses type device, the light emitter **1510** and the light receiver **1520** may be provided on the temple **191** of the AR device **1000**. For example, referring to FIG. **2**, the light emitter **1510** and the light receiver **1520** may be provided on an inner side part of the temple **191** of the AR device **1000**, which is a position between the temple **191** and user's eyes. For example, referring to FIG. **2**, the light emitter **1510** and the light receiver **1520** may be provided at positions spaced from the frame **110** by about 10 mm to 15 mm on the inner side of the temple **191** of the AR device **1000** The light emitter **1510** and the light receiver **1520** may be provided to face the light reflector **1400** in the temple **191** of the AR device **1000**.

Also, for example, referring to FIG. **2**, the light emitter **1510** and the light receiver **1520** may be provided on the nose support **192** of the AR device **1000**. The light emitter **1510** and the light receiver **1520** may be provided on an inner side part of the nose support **192** of the AR device **1000**, which is a position between the nose support **192** and the user's eyes. For example, referring to FIG. **2**, the light emitter **1510** and the light receiver **1520** may be provided at positions spaced from the frame **110** by about 10 mm to 15 mm on an inner side of the nose support **192** of the AR device **1000**. The light emitter **1510** and the light receiver **1520** may be provided to face the light reflector **1400** in the nose support **192** of the AR device **1000**.

The gaze detector **1500** may provide data related to the gaze of the user's eyes to the processor **1800**, and the processor **1800** may obtain gaze information of the user based on the data related to the gaze of the user's eyes. The data related to the gaze of the user's eyes is data obtained by the gaze detector **1500**, and may include a type (e.g., point light, line light, or planar light) of IR light emitted from the light emitter **1510**, characteristics of the IR light emitted from the light emitter **1510**, data regarding an emission region of the IR light emitted from the light emitter **1510**, and data indicating the characteristics of the IR light received from the light receiver **1520**. Further, the gaze information of the user is information related to the user's gaze, may be generated by analyzing the data related to the gaze of the user's eyes, and may include information about, for example, a location of the user's pupil, a location of a pupil central point, a location of the user's iris, a center of the user's eyes, a location of glint feature point of the user's eyes, a gaze point of the user, a gaze direction of the user, etc. but the disclosure is not limited thereto. The gaze direction of the user may be, for example, a direction of eth user's gaze from the center of the user's eyes toward the gaze point at which the user gazes. For example, the gaze direction of the user may be represented by a vector value from the center of the user's left eye toward the gaze point and a vector value from the center of the user's right eye toward the gaze point, but the disclosure is not limited thereto. According to an example embodiment of the disclosure, the gaze detector **1500** may detect data related to the gaze of the user wearing the AR device **1000** at a previously determined time interval.

The communication interface **1600** may transmit/receive data for receiving a service related to the AR device **1000** to/from an external device and a server.

The storage **1700** may store a program to be executed by the processor **1800**, which will be described later, and may store data input to or output from the AR device **1000**.

The storage **1700** may include at least one of an internal memory or an external memory. The internal memory may include, for example, at least one of a volatile memory (e.g., dynamic RAM (DRAM), static RAM (SRAM), synchronous dynamic RAM (SDRAM), etc.), a non-volatile memory (e.g., one time programmable ROM (OTPROM), programmable ROM (PROM), erasable and programmable ROM (EPROM), electrically erasable and programmable ROM (EEPROM), mask ROM, flash ROM, etc.), hard disk drive (HDD), or solid state drive (SSD). According to an example embodiment of the disclosure, the processor **1800** may load a command or data received from at least one of the non-volatile memory or another element into the volatile memory and process the command or the data. Also, the processor **1800** may store data received or generated from another element in the non-volatile memory. The external memory may include, for example, at least one of compact flash (CF), secure digital (SD), micro secure digital (Micro-SD), mini secure digital (Mini-SD), extreme digital (xD), or memory stick.

Programs stored in the storage **1700** may be classified into a plurality of modules according to their functions. According to an example embodiment, each of the plurality of modules may include one or more computer readable codes, which may include, for example, a light irradiation code **1710**, a light reception code **1720**, an eye feature detection code **1730**, a pattern detection code **1740**, a bias determination code **1750**, a pupil position detection code **1760**, a gaze determination code **1770**, and a calibration code **1780**. For example, a memory may be included in the gaze detector **1500**, and, in this case, the light irradiation code **1710** and the light reception code **1720** may be stored as firmware in the memory included in the gaze detector **1500**.

The processor **1800** controls the overall operation of the AR device **1000**. For example, the processor **1800** may execute the programs stored in the storage **1700**, thereby generally controlling the user inputter **1100**, the microphone **1200**, the display **1300**, the light reflector **1400**, the gaze detector **1500**, the communication interface **1600**, the storage **1700**, etc.

The processor **1800** may execute the light irradiation code **1710**, the light reception code **1720**, the eye feature detection code **1730**, the pattern detection code **1740**, the bias determination code **1750**, the pupil position detection code **1760**, the gaze determination code **1770**, and the calibration code **1780** that are stored in the storage **1700**, thereby determining the gaze point of the user and the gaze direction.

According to an example embodiment of the disclosure, the AR device **1000** may include a plurality of processors **1800**, and the light irradiation code **1710**, the light reception code **1720**, the eye feature detection code **1730**, the pattern detection code **1740**, the bias determination code **1750**, the pupil position detection code **1760**, the gaze determination code **1770**, and the calibration code **1780** may be executed by the plurality of processors **1800**.

For example, some of the light irradiation code **1710**, the light reception code **1720**, the eye feature detection code **1730**, the pattern detection code **1740**, the bias determination code **1750**, the pupil position detection code **1760**, the gaze determination code **1770**, and the calibration code **1780** may be executed by a first processor, and the others of the light irradiation code **1710**, the light reception code **1720**, the eye feature detection code **1730**, the pattern detection code **1740**, the bias determination code **1750**, the pupil position detection code **1760**, the gaze determination code **1770**, and the calibration code **1780** may be executed by a second processor, but the disclosure is not limited thereto.

For example, the gaze detector **1500** may include another processor and a memory, and the other processor may execute the light irradiation code **1710** and the light reception code **1720** that are stored in the memory, and the processor **1800** may execute the eye feature detection code **1730**, the pattern detection code **1740**, the bias determination code **1750**, the pupil position detection code **1760**, the gaze determination code **1770**, and the calibration code **1780** that are stored in the storage **1700**.

The processor **1800** may execute the light irradiation code **1710** stored in the storage **1700** so that the light emitter **1510** may emit IR light toward the light reflector **1400**. The processor **1800** may control the light emitter **1510** by executing the light irradiation code **1710**, and the light emitter **1510** controlled by the processor **1800** may emit the IR light toward at least a partial region of the light reflector **1400** so that the IR light reflected by the light reflector **1400** may cover the user's eyes.

For example, when the light receiver **1520** is an IR camera, the light emitter **1510** may be an IR LED, and the processor **1800** may control the IR LED so that the IR light emitted from the IR LED may be reflected by the light reflector **1400** and irradiated to a region including the user's eyes, in order for the IR camera to capture the user's eyes. For example, in order to reflect the light emitted from the IR LED by using the light reflector **1400** and irradiate the light to the region including the user's eyes, the processor **1800** may control an irradiation direction of the IR light emitted from the IR LED, and apply power to the IR LED, thereby controlling emission of the IR light from the IR LED.

According to one example embodiment of the disclosure, the IR camera and the IR LED may be installed toward the light reflector **1400** of the AR device **1000** so that the IR camera may capture the entire region of the user's eyes, and the processor **1800** may control the IR LED installed toward the light reflector **1400** to emit the IR light. An example in which the irradiation direction of the IR light emitted from the IR LED is controlled will be described in more detail with reference to FIG. **5**A.

According to another example embodiment of the disclosure, when the light receiver **1520** is an IR detector, the light emitter **1510** may be an IR scanner, and the processor **1800** may control the IR scanner to scan the user's eyes by reflecting the IR light emitted from the IR scanner by using the light reflector **1400**, so that the IR detector may detect the user's eyes. For example, in order to scan the user's eyes by reflecting the light emitted from the IR scanner by using the light reflector **1400**, the processor **1800** may control an irradiation direction of the IR light emitted from the IR scanner, and apply power to the IR scanner, thereby controlling emission of the IR light from the IR scanner. An example in which the irradiation direction of the IR light emitted from the IR scanner is controlled will be described in more detail with reference to FIGS. **5**B and **5**C.

The processor **1800** may execute the light reception code **1720** stored in the storage **1700** so that the light receiver **1520** may receive the light reflected by the light reflector **1400** from the user's eyes. The processor **1800** may control the light receiver **1520** by executing the light reception code

1720, and the light receiver **1520** controlled by the processor **1800** may receive the light reflected by the light reflector **1400** from the user's eyes.

For example, when the light emitter **1510** is an IR LED, the light receiver **1520** may be an IR camera, and the processor **1800** may control the IR camera to capture the user's eyes through the light reflected by the light reflector **1400** from the user's eyes.

Alternatively, for example, when the light emitter **1510** is an IR scanner, the light receiver **1520** may be an IR detector, and the processor **1800** may control the IR detector to detect the IR light reflected by the light reflector **1400** from the user's eyes, so that the IR detector may detect the user's eyes.

The processor **1800** may execute the eye feature detection code **1730** stored in the storage **1700**, thereby detecting features related to the gaze of the user's eyes. For example, the processor **1800** may execute the eye feature detection code **1730**, thereby detecting a position of a pupil feature point of the user's eyes and a position of a glint feature point of the user's eyes. The pupil feature point may be, for example, a pupil central point, and the glint feature point of the eyes may be a part having brightness greater than or equal to a certain value in a detected eye region. The position of the pupil feature point and the position of the glint feature point of the eyes may be identified, for example, by a coordinate value indicating a position in a coordinate system of the light receiver **1520**. For example, the coordinate system of the light receiver **1520** may be a coordinate system of an IR camera or a coordinate system of the IR detector, and the coordinate value in the coordinate system of the light receiver **1520** may be a 2D coordinate value.

The processor **1800** may detect features related to the gaze of the eyes by analyzing the light received by the light receiver **1520**. For example, when the light receiver **1520** is an IR camera, the processor **1800** may identify the position of the pupil feature point and the position of the glint feature point of the eyes in an image captured by the IR camera. Alternatively, for example, when the light receiver **1520** is an IR detector, the processor **1800** may analyze the IR light detected by the IR detector, thereby identifying the position of the pupil feature point and the position of the glint feature point of the eyes. When the positions of the feature points are identified based on the IR light detected by the IR detector, the position of the pupil feature point and the position of the glint feature point may have values calibrated by reflecting the bias of the support **190**. The position of the pupil feature point and the position of the glint feature point calibrated by reflecting the bias of the support **190** will be described in more detail with reference to FIG. **11**.

Also, the processor **1800** may analyze the light received by the light receiver **1520**, thereby obtaining a coordinate value indicating the position of the pupil feature point and a coordinate value indicating the position of the glint feature point of the eyes. For example, when the light receiver **1520** is an IR camera, the processor **1800** may obtain the coordinate value of the pupil feature point and the coordinate value of the glint feature point of the eyes from the coordinate system of the IR camera. The coordinate system of the IR camera may be used to indicate the position of the pupil feature point and the position of the glint feature point of the eyes, and, for example, coordinate values corresponding to pixels of an image captured by the IR camera on the coordinate system of the IR camera may be previously set. Also, based on a property (e.g., brightness) of IR light received through the IR camera, a coordinate value corresponding to a feature point of the eyes may be identified.

For example, when the light receiver **1520** is an IR camera, the processor **1800** may identify the position of the pupil central point in the image captured by the IR camera. The processor **1800** may identify the brightness of IR light received through an image sensor of the IR camera including a plurality of photodiodes, and identify at least one pixel that receives IR light indicating the pupil among the pixels of the image captured by the IR camera, thereby identifying the position of the pupil central point. For example, positions of the pixels in the image captured by an IR camera may be identified through the coordinate system of the IR camera, and the position of the pupil central point may have a coordinate value in the coordinate system of the IR camera, as a position value of at least one pixel corresponding to the pupil central point.

For example, the processor **1800** may identify a position of the brightest point in the image captured by the IR camera, in order to identify the glint feature point of the eyes. The processor **1800** may identify the brightness of the IR light received through the image sensor of the IR camera including the plurality of photodiodes, and may identify at least one pixel corresponding to bright IR light equal to or greater than a certain reference among the pixels of the image captured by the IR camera, thereby identifying the position of the glint feature point of the eyes. For example, the processor **1800** may identify the pixel corresponding to the brightest IR light among the pixels of the image captured by the IR camera, thereby identifying the position of the glint feature point of the eyes. For example, the positions of the pixels in the image captured by the IR camera may be identified through the coordinate system of the IR camera, and the position of the glint feature point of the eyes may have the coordinate value in the coordinate system of the IR camera, as a position value of the pixel corresponding to the glint feature point.

Alternatively, for example, when the light receiver **1520** is an IR detector, the processor **1800** may calculate the coordinate value of the pupil feature point and the coordinate value of the glint feature point of the eyes in the coordinate system of the IR detector.

When the light emitter **1510** is an IR scanner, the processor **1800** may control the IR scanner to sequentially irradiate a point light source ora line light source to cover a region where the user's eyes are located, and sequentially receive the light reflected from the user's eyes through the IR detector in order to scan the region where the user's eyes are located. In addition, the processor **1800** may analyze an array of light sequentially received through the IR detector, thereby identifying the pupil feature point and the glint feature point of the eyes.

The coordinate system of the IR detector may be used to indicate the position of the pupil feature point of the pupil and the position of the glint feature point of the eyes, and, for example, coordinate values corresponding to the lights in the array of lights sequentially received through the IR detector on the coordinate system of the IR detector may be previously set. For example, irradiation directions and irradiation times of lights emitted from the IR scanner may be determined according to an operation setting value of the IR scanner, and a light array may be formed from the lights emitted from the IR scanner. For example, based on the irradiation direction and irradiation time of the lights emitted from the IR scanner, and reception time of the lights received from the IR detector, coordinate values corresponding to the lights in the light array on the coordinate system of the IR detector may be identified. In addition, based on the property (e.g., brightness) of the lights in the array of lights sequentially received through the IR detector, light corresponding to the feature point of the eye and coordinate values of the light may be identified.

For example, the processor **1800** may identify lights having a brightness equal to or less than a certain value in the received light array, thereby identifying the position of the pupil feature point based on coordinate values corresponding to the identified lights on the coordinate system of the IR detector.

For example, the processor **1800** may identify light having a brightness equal to or greater than a certain value in the received light array, thereby identifying a coordinate value corresponding to the identified light on the coordinate system of the IR detector as the coordinate value of the glint feature point of the eyes.

Also, for example, when the light receiver **1520** is an IR detector, the coordinate value of the pupil feature point and the coordinate value of the glint feature point of the eyes may be values calibrated by reflecting the degree of bias of the support **190** of the AR device **1000**, which will be described below. In this case, the processor **1800** may calculate the coordinate value of the pupil feature point and the coordinate value of the glint feature point of the eyes calibrated by reflecting the degree of bias of the temple **191** of the AR device **1000** and/or the degree of bias of the nose support **192**. The calibrated coordinate values may be input to a mapping function which will be described below.

The processor **1800** may execute the pattern detection code **1740** stored in the storage **1700**, thereby detecting a pattern of the light reflector **1400**. The light reflector **1400** may be coated on one surface of the waveguide **170** of the AR device **1000** to have a certain pattern. The processor **1800** may receive the IR light reflected by the user's eyes and reflected by the light reflector **1400** through the light receiver **1520**, and identify a shape of the pattern based on the received IR light. The pattern formed on the light reflector **1400** may include, for example, a dot pattern, a line pattern, a grid pattern, a 2D marker, etc., but the disclosure is not limited thereto. An example of the pattern formed on the light reflector **1400** and identified by the pattern detection code **1740** will be described in more detail with reference to FIGS. **7**A to **7**D. When the temple **191** is biased with respect to the frame **110**, the pattern identified by the pattern detection code **1740** may have a deformed shape.

For example, when the light receiver **1520** is an IR camera, the IR camera may capture the user's eyes based on the IR light reflected by the light reflector **1400**, and the processor **1800** may identify a pattern within an image obtained by capturing the user's eyes from the image.

For example, when the light receiver **1520** is an IR detector, the IR detector may sequentially receive IR lights reflected by the light reflector **1400**, and the processor **1800** may identify a part related to the pattern of the light reflector **1400** in an array of the sequentially received IR lights.

The processor **1800** may execute the bias determination code **1750** stored in the storage **1700**, thereby determining a degree to which the support **190** of the AR device **1000** is biased with respect to the frame **110**. The support **190** may include, for example, the temple **191** and the nose support **192**. When the user wears the AR device **1000**, the temple **191** may be widened or narrowed according to the size of the user's head and face. At this time, when the IR light is received after the support **190** is biased with respect to the frame **110**, the bias determination code **1750** may identify a pattern having a deformed shape from the received IR light. Also, the bias determination code **1750** may compare the pattern having the deformed shape with a non-deformed pattern, thereby estimating the degree of bias of the support **190**. For example, the bias determination code **1750** may compare the pattern having the deformed shape with the non-deformed pattern to identify degree of deformation of the pattern, and may determine degree of bias of the support **190** based on the degree of deformation of the pattern. Alternatively, the bias determination code **1750** may analyze only the pattern having the deformed shape without comparing the pattern having the deformed shape with the non-deformed pattern, thereby estimating the degree of bias of the support **190**. For example, when the pattern is a dot pattern, the bias determination code **1750** may identify a difference between intervals between points in the pattern having the deformed shape, thereby estimating the degree of bias of the support **190**.

For example, the degree of bias of the temple **191** may be expressed as a bias angle indicating a difference between a default angle of the temple **191** with respect to the frame **110** and an angle of the biased temple **191** with respect to the frame **110**, but the disclosure is not limited thereto. Also, for example, the degree of bias of the nose support **192** may be expressed as a bias angle indicating a difference between a default angle of the nose support **192** with respect to the frame **110** and an angle of the biased nose support **192** with respect to the frame **110**, but the disclosure is not limited thereto.

In the above, it has been described that the pattern detection code **1740** detects the deformation of the pattern formed on the light reflector **1400**, and the bias determination code **1750** analyzes the pattern having the deformed shape to estimate the degree of bias of the support **190**, but the disclosure is not limited thereto.

For example, a certain pattern may be formed on a part of the AR device **1000** that may reflect light emitted from the light emitter **1510** to direct the reflected light toward the light receiver **1520**. For example, the pattern may be formed on a partial region of the frame **110** of the AR device **1000** directed toward the light emitter **1510** and the light receiver **1520**. For example, the pattern may be formed by forming a certain curve in a partial region of the frame **110**. Alternatively, for example, the pattern may be formed by forming a material capable of reflecting light on a partial region of the frame **110**. In addition, the pattern may be attached to or coated on other elements included in the glasses type AR device **1000**, for example, a vision correcting lens for vision correction or a cover glass installed to protect a waveguide.

Alternatively, for example, the pattern may be formed on a partial region of the nose support **192** of the AR device **1000** directed toward the light emitter **1510** and the light receiver **1520**. For example, the pattern may be formed by forming a certain curve in a partial region of the nose support **192**. Alternatively, for example, the pattern may be formed by forming a material capable of reflecting light on a partial region of the nose support **192**. In this case, the light emitter **1510** and the light receiver **1520** are preferably located in the temple **191** of the AR device **1000**, but the disclosure is not limited thereto.

Alternatively, for example, the pattern may be formed in a partial region of the temple **191** of the AR device **1000** directed toward the light emitter **1510** and the light receiver **1520**. For example, the pattern may be formed by forming a certain curve in a partial region of the temple **191**. Alternatively, for example, the pattern may be formed by forming a material capable of reflecting light on a partial region of the temple **191**. In this case, the light emitter **1510** and the light receiver **1520** are preferably located on the nose support **192** of the AR device **1000**, but the disclosure is not limited thereto.

The processor **1800** may execute the pupil position detection code **1760** stored in the storage **1700**, thereby detecting a pupil position of the user's eyes. The pupil position detection code **1760** may identify the pupil position of the user's eyes based on the IR light reflected from the light reflector **1400**.

For example, when the light receiver **1520** is an IR camera, the pupil position detection code **1760** may identify the pupil position of the user's eyes within an image captured by the IR camera from the image. Alternatively, for example, when the light receiver **1520** is an IR detector, the pupil position detection code **1760** may analyze the IR light sequentially obtained by the IR detector, thereby calculating the pupil position of the user's eyes.

The pupil position detection code **1760** may identify the pupil central point of the user's eyes, thereby identifying the pupil position of the user's eyes.

The processor **1800** may execute the gaze determination code **1770** stored in the storage **1700**, thereby obtaining information about the user's gaze. The processor **1800** may execute the gaze determination code **1770**, thereby calculating a position of the center of the user's eyes. The center of the user's eyes may be the center of user's eyeballs. The processor **1800** may calculate the position of the center of the user's eyes, based on the pupil position of the user's eyes obtained by the pupil position detection code **1760** and the degree of bias of the support **190** obtained by the bias determination code **1750**. For example, the processor **1800** may calculate the position of the center of the user's eyes so that a value calculated based on a matrix for calibrating the degree of bias of the support **190**, a value indicating the position of the center of the user's eyes and a bias of axis of the image obtained by capturing the user's eyes can be a value of the pupil position of the user's eyes obtained by the pupil position detection code **1760**. For example, the center of the eye may be the center of the eyeball, and the position of the center of the user's eyes may have a 3D coordinate value in a coordinate system of a real space.

The processor **1800** may execute the gaze determination code **1770**, thereby calculating a position of the gaze point of the user. In order to calculate the position of the gaze point of the user, the processor **1800** may previously generate a mapping function for calculating the position of the gaze point from features of the user's eyes. The mapping function is a function for calculating the position of the gaze point of the user in consideration of features of the user's eyes and bias information of the support **190**, and may be generated during a calibration process of the calibration code **1780** which will be described below. For example, the position of the gaze point may have a 3D coordinate value in the coordinate system in the real space, but the disclosure is not limited thereto. For example, the position of the gaze point may have a coordinate value in the coordinate system of the waveguide **170**, but the disclosure is not limited thereto.

The processor **1800** may execute the gaze determination code **1770**, thereby calibrating the features related to the user's gaze obtained from the eye feature detection code **1730** based on the degree of bias obtained from the bias determination code **1750**. Also, the processor **1800** may apply the features related to the user's gaze calibrated based on the degree of bias to the mapping function, thereby calculating the position of the gaze point of the user. Also, a gaze direction of the user may be determined based on the position of the central point of the user's eyes and the gaze point of the user calculated by the gaze determination code **1770**. A method of obtaining the gaze direction of the user by using the gaze determination code **1770** will be described in more detail with reference to FIG. **11**.

Alternatively, the processor **1800** may calculate the gaze point of the user without using the above-described mapping function. For example, when the light receiver **1520** is an IR camera, the gaze determination code **1770** may calculate the gaze direction of the user's eyes from the image obtained by capturing the user's eyes by using a certain algorithm. In this case, the obtained gaze direction may be a vector value indicating the gaze direction of the user's eyes in the camera coordinate system. The algorithm used to obtain the gaze direction of the user's eyes may be an algorithm for fitting a 3D eye model. The algorithm for fitting the 3D eye model may be an algorithm for obtaining a vector value indicating the gaze direction of the user by comparing an eye image corresponding to a reference vector value indicating the gaze direction of the user with an image captured by an IR camera In addition, the gaze determination code **1770** may convert the vector value indicating the gaze direction in the camera coordinate system into a vector value indicating the gaze direction in the coordinate system of the waveguide**170** by using the bias angle of the temple **191**. Thereafter, the gaze determination code **1770** may calculate an intersection point between a vector indicating the gaze direction in the coordinate system of the waveguide **170** and the waveguide **170**, thereby obtaining the gaze point of the user.

The processor **1800** may execute the calibration code **1780** stored in the storage **1700**, thereby calibrating the mapping function based on the bias angle of the support **190**. The processor **1800** may execute the calibration code **1780**, thereby calibrating the mapping function to obtain the gaze point of the user based on a previously set default bias angle and features of eyes.

For example, when the light receiver **1520** is an IR camera, the processor **1800** may display a target point for calibration through the waveguide**170**, and capture the user's eyes looking at the target point by using the IR camera. In addition, the processor **1800** may identify and analyze a pattern within the image obtained by capturing the user's eyes, thereby obtaining a bias angle of the support **190**. In addition, the processor **1800** may detect positions of the feature points related to the user's eyes from the image obtained by capturing the user's eyes, and input the positions of the feature points of the user's eyes and the bias angle of the support **190** into the mapping function. The processor **1800** may calibrate the mapping function so that a position value of the target point may be output from the mapping function to which the positions of the feature points of the user's eyes and the bias angle of the support **190** are input.

For example, when the light receiver **1520** is an IR detector, the processor **1800** may display the target point for calibration through the waveguide**170** and control the IR scanner to emit IR light for scanning the user's eyes looking at the target point. In addition, the processor **1800** may receive and analyze IR light reflected from the user's eyes through the IR detector, identify a pattern of the light reflector **1400**, and estimate the bias angle of the temple **191**. The processor **1800** may analyze the IR light based on the bias angle of the temple **191**, thereby identifying positions of the calibrated feature points of eyes. For example, when the processor **1800** estimates the positions of the feature points of eyes by using the IR scanner and the IR detector, because results of estimating the positions of the feature points of eyes are affected by an operating angle of the IR scanner, the positions of the feature points of eyes may be calibrated based on the bias of the support **190** by calculating a value obtained by subtracting the bias angle of the support **190** from the operating angle of the IR scanner.

In addition, the processor **1800** may input the positions of the calibrated feature points of the eyes into the mapping function, and calibrate the mapping function so that the position value of the target point may be output from the mapping function in which the calibrated positions of the feature points of the eyes are input.

FIG. **4** is a diagram illustrating an example of operations of the light emitter **1510** and the light receiver **1520** of the AR device **1000** according to an example embodiment of the disclosure.

Referring to FIG. **4**, the light emitter **1510** may emit IR light toward the light reflector **1400**, and the emitted IR light may be reflected by the light reflector **1400** and directed toward user's eyes. In addition, the IR light directed toward the user's eyes may be reflected back by the user's eyes and directed toward the light reflector **1400**, and the IR light reflected by the user's eyes may be reflected back by the light reflector **1400** and directed toward the light receiver **1520**. Also, the light receiver **1520** may receive IR light reflected from the user's eyes by the light reflector **1400** and directed toward the light receiver **1520**.

In FIG. **4**, for convenience of explanation, it has been described that the AR device **1000** which is a glasses type display device emits IR light toward the user's left eye and receives reflected IR light from the user's left eye, but the disclosure is not limited thereto. The AR device **1000** may emit IR light toward the user's right eye and receive the reflected IR light from the user's right eye in the same manner as shown in FIG. **4**.

FIG. **5**A is a diagram illustrating an example of the light emitter **1510** that emits planar light according to an example embodiment of the disclosure.

Referring to FIG. **5**A, the light emitter **1510** of FIG. **2** may be an IR LED, and the light receiver **1520** of FIG. **2** may be an IR camera. In this case, the light emitter **1510** may emit IR light of the planar light toward the light reflector **1400**, and the emitted IR light may be reflected by the light reflector **1400** and directed toward a user's eye. For example, in order to reflect the light emitted from the IR LED by using the light reflector **1400** and irradiate the light to the region including the user's eye, the processor **1800** may control an irradiation direction of the IR light emitted from the IR LED, and apply power to the IR LED, thereby controlling emission of the IR light from the IR LED. In addition, the IR light of the planar light reflected by the light reflector **1400** may cover the entire user's eye. In this case, the light receiver **1520** may be an IR camera, and the IR camera may receive the IR light reflected by the user's eye, thereby capturing the user's eye.

For example, coordinate values corresponding to pixels of an image captured by the IR camera on the coordinate system of the IR camera may be previously set. In addition, the processor **1800** may identify a coordinate value corresponding to a feature point of the eye based on a property (e.g., brightness) of IR light received through the IR camera. For example, the processor **1800** may identify the brightness of IR light received through an image sensor of the IR camera including a plurality of photodiodes, and identify at least one pixel that receives IR light indicating the pupil among the pixels of the image captured by the IR camera, thereby identifying a coordinate value **51** corresponding to the pupil central point on the IR coordinate system. In this case, the IR light representing the pupil may be an IR light having a brightness lower than a certain value, but the disclosure is not limited thereto.

Also, for example, the processor **1800** may identify the brightness of the IR light received through the image sensor of the IR camera including the plurality of photodiodes, and may identify at least one pixel representing the glint feature point of the eye among the pixels of the image captured by the IR camera, thereby identifying a coordinate value **52** corresponding to a glint feature point on the IR coordinate system. In this case, the IR light representing the glint feature point of the eye may be an IR light having a brightness greater than a certain value, but the disclosure is not limited thereto.

According to an example embodiment of the disclosure, the light emitter **1510** may be an IR LED that emits blinking light, and in this case, the light reflector **1400** may be an IR event camera. The IR event camera may be an IR camera that is activated when a specific event occurs and automatically captures a subject. The IR event camera may be activated to automatically capture the user's eye, for example, when patterns of blinking light are different.

FIG. **5**B is a diagram illustrating an example of the light emitter **1510** that emits point light according to an example embodiment of the disclosure.

Referring to FIG. **5**B, the light emitter **1510** of FIG. **2** may be an IR scanner, and the light receiver **1520** of FIG. **2** may be an IR detector. In this case, the light emitter **1510** may emit IR light of a point light toward the light reflector **1400**, and the emitted IR light may be reflected by the light reflector **1400** and directed toward user's eye. In this case, the light emitter **1510** may sequentially emit IR lights of point light toward the light reflector **1400** while changing an emission direction to a vertical direction or a horizontal direction, and the sequentially emitted IR lights of point light may be reflected by the light reflector **1400** to cover the entire user's eye. For example, in order to reflect the IR lights of point light sequentially emitted from the IR scanner by using the light reflector **1400** and irradiate the IR lights of point light to a region including the user's eye, the processor **1800** may control irradiation directions of the IR lights emitted from the IR scanner. In this case, the light emitter **1510** may be a 2-dimensional (2D) scanner and the light receiver **1520** may be at least one photodiode.

For example, coordinate values corresponding to the IR lights in the array of lights sequentially received through the IR detector on the coordinate system of the IR detector may be previously set, and the processor **1800** may identify a coordinate value corresponding to a feature point of the eye based on properties (e.g., brightness) of the IR lights in the array of IR lights sequentially received through the IR detector. For example, the processor **1800** may identify coordinate values corresponding to IR lights having a brightness equal to or less than a certain value in the array of the received IR lights on the coordinate system of the IR detector, thereby identifying a coordinate value **53** corresponding to the pupil center point on the IR coordinate system. Also, for example, the processor **1800** may identify coordinate values corresponding to IR lights having a brightness equal to or less than a certain value in the array of the received IR lights on the coordinate system of the IR detector, thereby identifying a coordinate value **54** corresponding to the glint feature point of the eye on the IR coordinate system.

FIG. **5**C is a diagram illustrating an example of the light emitter **1510** that emits line light according to an example embodiment of the disclosure.

Referring to FIG. **5**C, the light emitter **1510** of FIG. **2** may be an IR scanner, and the light receiver **1520** of FIG. **2** may be an IR detector. In this case, the light emitter **1510** may emit IR light of line light toward the light reflector **1400**, and the emitted IR light may be reflected by the light reflector **1400** and directed toward user's eye. For example, in order to reflect the IR lights of line light sequentially emitted from the IR scanner by using the light reflector **1400** and irradiate the IR lights of point light to a region including the user's eye, the processor **1800** may control irradiation directions of the IR lights emitted from the IR scanner. In this case, the light emitter **1510** may sequentially emit IR lights of line light toward the light reflector **1400** while changing an emission direction, and the sequentially emitted IR lights of line light may be reflected by the light reflector **1400** to cover the entire user's eye. In this case, the light emitter **1510** may be a 1-dimensional (1D) scanner and the light receiver **1520** may be a photodiode array including a plurality of photodiodes. When the light receiver **1520** is the photodiode array, it is more preferable that the light receiver **1520** is provided in the temple **191**. For example, when horizontal line light is emitted from the light emitter **1510**, the light emitter **1510** may change the emission direction to a vertical direction to cover the region including the user's eye, and, when a vertical line light is emitted from the light emitter **1510**, the light emitter **1510** may change the emission direction to the horizontal direction.

For example, coordinate values corresponding to the IR lights in the array of lights sequentially received through the IR detector on the coordinate system of the IR detector may be previously set, and the processor **1800** may identify a coordinate value corresponding to a feature point of the eye based on properties (e.g., brightness) of the IR lights in the array of IR lights sequentially received through the IR detector. For example, the processor **1800** may identify line lights of IR light having a brightness greater than or equal to a certain value, and identify a photodiode corresponding to IR light having a brightness less than or equal to the certain value among a plurality of photodiodes that have received the line lights according to the respective line lights, thereby identifying a coordinate value **55** corresponding to the pupil center point on the IR coordinate system. In addition, for example, the processor **1800** may identify line lights of IR light having a brightness greater than or equal to a certain value, and identify a photodiode corresponding to IR light having a brightness less than or equal to the certain value among a plurality of photodiodes that have received the line lights according to the respective line lights, thereby identifying a coordinate value **56** corresponding to the glint feature point of the eye on the IR coordinate system.

In FIGS. **5**A to **5**C, for convenience of description, an example in which the AR device **1000** which is the glasses type display device, emits IR light toward user's one eye has been described, but the disclosure is not limited thereto. The AR device **1000** may emit IR light toward the user's other eye in the same manner as illustrated in FIGS. **5**A to **5**C.

FIG. **6**A is a diagram illustrating an example in which the light emitter **1510** and the light receiver **1520** are provided in the temple **191** of the AR device **1000** according to an example embodiment of the disclosure.

Referring to FIG. **6**A, the AR device **1000** of FIG. **2** may be a glasses type device, and the light emitter **1510** and the light receiver **1520** may be provided on the temple **191** of the AR device **1000**. The light emitter **1510** and the light receiver **1520** may be provided on an inner side part of the temple **191** of the AR device **1000**, which is a position between the temple **191** and user's eyes. For example, the light emitter **1510** and the light receiver **1520** may be provided at positions spaced from a frame by about 10 mm to 15 mm on an inner side of the temple **191** of the AR device **1000** The light emitter **1510** and the light receiver **1520** may be provided to face the light reflector **1400** in the temple **191** of the AR device **1000**.

FIG. **6**B is a diagram illustrating an example in which the light emitter **1510** and the light receiver **1520** are provided in the nose support **192** of the AR device **1000** according to an example embodiment of the disclosure.

Referring to FIG. **6**B, the AR device **1000** of FIG. **2** may be a glasses type device, and the light emitter **1510** and the light receiver **1520** may be provided on the nose support **192** of the AR device **1000**. The light emitter **1510** and the light receiver **1520** may be provided on an inner side part of the nose support **192** of the AR device **1000**, which is a position between the nose support **192** and user's eyes. The light emitter **1510** and the light receiver **1520** may be provided to face the light reflector **1400** in the nose support **192** of the AR device **1000**.

FIG. **6**C is a diagram illustrating an example in which the light emitter **1510** and the light receiver **1520** are provided in the temple **191** and the nose support **192** of the AR device **1000** according to an example embodiment of the disclosure.

Referring to FIG. **6**C, the AR device **1000** of FIG. **2** may be a glasses type device, the light emitter **1510** may be provided on the temple **191** of the AR device **1000**, and the light receiver **1520** may be provided on the nose support **192** of the AR device **1000**. The light emitter **1510** may be provided on an inner side part of the temple **191** of the AR device **1000**, which is a position between the temple **191** and user's eyes. The light receiver **1520** may be provided on an inner side part of the nose support **192** of the AR device **1000**, which is a position between the nose support **192** and user's eyes. The light emitter **1510** and the light receiver **1520** may be provided to face the light reflector **1400** in the AR device **1000**.

FIG. **6**D is a diagram illustrating an example in which the light emitter **1510** and the light receiver **1520** are provided in the temple **191** and the nose support **192** of the AR device **1000** according to an example embodiment of the disclosure.

Referring to FIG. **6**D, the AR device **1000** of FIG. **2** may be a glasses-type device, the light emitter **1510** may be provided on the nose support **192** of the AR device **1000**, and the light receiver **1520** may be provided on the temple **191** of the AR device **1000**. The light emitter **1510** may be provided on an inner side part of the nose support **192** of the AR device **1000**, which is a position between the nose support **192** and user's eyes. The light receiver **1520** may be provided on an inner side part of the temple **191** of the AR device **1000**, which is a position between the temple **191** and user's eyes. The light emitter **1510** and the light receiver **1520** may be provided to face the light reflector **1400** in the AR device **1000**.

In FIGS. **6**A to **6**D, it has been described that one light emitter **1510** and one light receiver **1520** are provided in the AR device **1000**, but the disclosure is not limited thereto. For example, a plurality of light emitters **1510** may be provided in the AR device **1000**. In this case, the plurality of light emitters **1510** may be provided on the temple **191**, or the plurality of light emitters **1510** may be provided on the nose support **192**. Alternatively, the plurality of light emitters **1510** may be dividedly provided on the temple **191** and the nose support **192**.

Also, for example, the plurality of light receivers **1520** may be provided in the AR device **1000**. In this case, the plurality of light receivers **1520** may be provided on the temple **191**, or the plurality of light receivers **1520** may be provided on the nose support **192**. Alternatively, the plurality of light receivers **1520** may be dividedly provided on the temple **191** and the nose support **192**.

In FIGS. **6**A to **6**D, for convenience of description, it has been described that the light emitter **1510** and the light receiver **1520** are provided in a left eye part of the AR device **1000**, which is the glasses type display device, but the disclosure is not limited thereto. In the AR device **1000**, the light emitter **1510** and the light receiver **1520** may be provided in a right eye part of the AR device **1000** in the same manner as illustrated in FIGS. **6**A to **6**D.

FIG. **7**A is a diagram illustrating an example of a dot pattern formed on the light reflector **1400** of the AR device **1000** according to an example embodiment of the disclosure, FIG. **7**B is a diagram illustrating an example of a grid pattern formed on the light reflector **1400** of the AR device **1000** according to an example embodiment of the disclosure, FIG. **7**C is a diagram illustrating an example of a pattern in the form of a 2D marker according to an example embodiment of the disclosure, and FIG. **7**D is a diagram illustrating an example of the light reflector **1400** that covers a part of the waveguide **170** according to an example embodiment of the disclosure.

Referring to FIG. **7**A, the dot-shaped pattern may be formed on the light reflector **1400** of the AR device **1000** of FIG. **2**. Referring to FIG. **7**B, the grid-shaped pattern may be formed on the light reflector **1400** of the AR device **1000** of FIG. **2**. According to an example embodiment, IR light may not be reflected from a part where the pattern is formed. At this time, because the dot-shaped pattern or the grid-shaped pattern is for detecting bias of a support or a temple of an AR device, it is preferable to have a regular shape.

Referring to FIG. **7**C, the pattern formed on a part of the light reflector **1400** of the AR device **1000** of FIG. **2** at which the user's gaze is less frequently directed. The formed pattern may be, for example, a pattern in the form of the 2D marker, but the disclosure is not limited thereto. The pattern may be formed on, for example, a part of the light reflector **1400** that does not interfere with capturing or scanning the user's eyes.

Referring to FIG. **7**D, the light reflector **1400** may be formed on a part of the waveguide **170** of the AR device **1000** of FIG. **2**. For example, the light reflector **1400** may not be located in a part of the waveguide **170** which has little relation to reflection of IR light.

For example, when the light receiver **1520** is an IR camera, the IR camera may capture the user's eyes based on the IR light reflected by the light reflector **1400**, and the IR light may not be reflected from the part where the pattern is formed. For example, a part of an image obtained by capturing the user's eyes in which the IR light is not reflected may be in black, and the processor **1800** may identify the black part in the image obtained by capturing the user's eyes, thereby identifying the pattern in the image.

For example, when the light receiver **1520** is an IR detector, the IR detector may sequentially receive IR lights reflected by the light reflector **1400**, and the IR light may not be reflected from the part where the pattern is formed. For example, the processor **1800** may identify a part of an IR light array formed by the sequentially received IR lights, from which the IR light is not reflected, thereby identifying the pattern of the light reflector **1400**.

FIG. **8**A is a diagram illustrating a light emission angle and a pattern before the temple **191** of the AR device **1000** is biased according to an example embodiment of the disclosure, and FIG. **8**B is a diagram illustrating a light emission angle and a pattern after the temple **191** of the AR device **1000** is biased according to an example embodiment of the disclosure.

Referring to FIG. **8**A, in the glasses type AR device **1000** as shown in FIG. **2**, before the temple **191** is biased, the light emitter **1510** may emit IR light toward the light reflector **1400** on which a point pattern is formed. According to an example embodiment, the light emitter **1510** may emit IR light toward the light reflector **1400** at a first light emission angle **80**, and a first pattern **82** may be identified by the AR device **1000** based on the IR light received by the light receiver **1520**.

In addition, referring to FIG. **8**B, in the glasses type AR device **1000** as shown in FIG. **2**, after the temple **191** is biased, the light emitter **1510** may emit IR light toward the light reflector **1400** at a second light emission angle **90**, and a second pattern **92** may be identified by the AR device **1000** based on the IR light received by the light receiver **1520**.

As shown in FIGS. **8**A and **8**B, the first pattern **82** and the second pattern **92** may include dots with different spaces from each other, and the AR device **1000** may compare the first pattern **82** and the second pattern **92**, thereby identifying degree of bias of the temple **191**. For example, a bias angle with respect to the first pattern **82** may be set to '0', and a bias angle with respect to the second pattern **92** may be calculated based on differences in positions between points in the first pattern **82** and points in the second pattern **92**. For example, the bias angle with respect to the second pattern **92** may be a difference value between the first light emission angle **80** and the second light emission angle **90**. For example, when a pattern corresponding to the first light emission angle **80** is the first pattern **82** and a pattern corresponding to the first light emission angle **80** is the second pattern **92**, the processor **1800** may compare spaces of the dots in the first pattern **82** with the spaces of the dots in the second pattern **92**, thereby identifying difference values of the spaces of the dots in the second pattern **92** with respect to the spaces of the dots in the first pattern **82**, and may input the identified difference values into at least one function for calculating degree of bias of the temple **191**, thereby calculating the degree of bias of the temple **191** indicating a difference between the first light emission angle **80** corresponding to the first pattern **82** and the second light emission angle **90** corresponding to the second pattern **92**.

In addition, for example, when a pattern corresponding to the first light emission angle **80** is the first pattern **82** and a pattern corresponding to the first light emission angle **80** is the second pattern **92**, the processor **1800** may compare positions of the dots in the first pattern **82** with the positions of the dots in the second pattern **92**, thereby identifying difference values of the positions of the dots in the second pattern **92** with respect to the positions of the dots in the first pattern **82**, and may input the identified difference values into at least one function for calculating degree of bias of the temple **191**, thereby calculating the degree of bias of the temple **191** indicating a difference between the first light emission angle **80** corresponding to the first pattern **82** and the second light emission angle **90** corresponding to the second pattern **92**.

According to another example embodiment, the processor **1800** may identify the degree of bias of the temple **191** based on the second pattern **92** without comparing the first pattern **82** and the second pattern **92**. In this case, the processor **1800** may identify the difference values of the spaces of the dots in the second pattern **92**, and input the identified difference values into at least one function for calculating the degree of bias of the temple **191**, thereby calculating the degree of bias of the temple **191**.

FIG. **9** is a diagram illustrating an example of a pattern identified from an array of light received through the light receiver **1520** when a light emitter of the AR device **1000** is an IR scanner **1520-1** or **1520-2** according to an example embodiment of the disclosure.

Referring to FIG. **9**, the IR scanner **1520-1** may represent an IR scanner before the support **190** of the glasses type AR device **1000** as shown in FIG. **2** is biased. The IR scanner **1520-1** may sequentially emit IR light of point lint toward a region where user's eyes are located by using the light reflector **1400**, and the light receiver **1520** may receive a first light array **90**.

In addition, the IR scanner **1520-2** may represent the IR scanner after the support **190** of the AR device **1000** is biased. The IR scanner **1520-2** may sequentially emit IR light of point lint toward the region where user's eyes are located by using the light reflector **1400**, and the light receiver **1520** may receive a second light array **92**.

In the first light array **90** and the second light array **92**, positions and spaces of light signals corresponding to dot patterns of the light reflector **1400** may be different from each other, and the AR device **1000** may compare a part corresponding to the dot pattern of the light reflector **1400** in the first light array **90** and a part corresponding to the dot pattern of the light reflector **1400** in the second light array **92**. Also, the AR device **1000** may identify degree of bias of the support **190** of the AR device **1000** based on results of comparison.

For example, because IR light emitted toward the dot pattern is not reflected by the light reflector **1400**, light signal may not be received in parts **90-1**, **90-2**, **90-3**, and **90-4** of the first light array **90** corresponding to dot patterns and parts **92-1**, **92-2**, **92-3**, and **92-4** of the second light array **92** corresponding to dot patterns. The processor **1800** may identify parts of the first light array **90** from which no light signal is received, thereby identifying the parts **90-1**, **90-2**, **90-3**, and **90-4** of the first light array **90** corresponding to the dot patterns, and identifying coordinate values **90-5**, **90-6**, **90-7**, and **90-8** on a coordinate system of the IR scanner **1520-1** respectively indicating the parts **90-1**, **90-2**, **90-3**, and **90-4** corresponding to the dot patterns. Also, for example, the processor **1800** may identify parts of the second light array **92** from which no light signal is received, thereby identifying the parts **92-1**, **92-2**, **92-3**, and **92-4**of the second light array **92** corresponding to the dot patterns, and identifying coordinate values **92-5**, **92-6**, **92-7**, and **92-8** on a coordinate system of the IR scanner **1520-2** respectively indicating the parts **92-1**, **92-2**, **92-3**, and **92-4** corresponding to the dot patterns.

The processor **1800** may compare the coordinate values **90-5**, **90-6**, **90-7**, and **90-8** on the coordinate system of the IR scanner **1520-1** respectively indicating the parts **90-1**, **90-2**, **90-3**, and **90-4** of the first light array **90** corresponding to the dot patterns with the coordinate values **92-5**, **92-6**, **92-7**, and **92-8** on the coordinate system of the IR scanner **1520-2** respectively indicating the parts **92-1**, **92-2**, **92-3**, and **92-4** of the second light array **92** corresponding to the dot patterns, thereby identifying differences between the coordinate values **90-5**, **90-6**, **90-7**, and **90-8** and the coordinate values **92-5**, **92-6**, **92-7**, and **92-8**. For example, the processor **1800** may calculate the degree of bias of the support **190** of the AR device **1000**, based on a difference value between the coordinate value **90-5** and the coordinate value **92-5**, a difference value between the coordinate value **90-6** and the coordinate value **92-6**, a difference value between the coordinate value **90-7** and the coordinate value **92-7**, and a difference value between the coordinate value **90-8** and the coordinate value **92-8**.

In FIG. **9**, for convenience of description, the IR scanner **1520-1** before the support **190** is biased and the IR scanner **1520-2** after the support **190** of the AR device **1000** is biased are illustrated separately in FIG. **9**, in order to distinguish the IR scanner **1520-1** before the support **190** of the AR device **1000** is biased from the IR scanner **1520-2** after the support **190** of the AR device **1000** is biased. However, the IR scanner **1520-1** before the support **190** is biased and the IR scanner **1520-2** after the support **190** of the AR device **1000** may be a same IR scanner installed in the AR device **1000**.

In addition, in FIG. **9**, for convenience of description, it has been described that one first light array **90** is used to identify the dot pattern using the IR scanner **1520-1** before the support **190** is biased, and one second light array **92** is used to identify the dot pattern using the IR scanner **1520-2** after the support **190** of the AR device **1000** is biased, but the disclosure is not limited thereto. A plurality of light arrays for covering the dot pattern may be used to identify the dot pattern using the IR scanner **1520-1** before the support **190** is biased, and a plurality of light arrays for covering the dot pattern may be used to identify the dot pattern using the IR scanner **1520-2** after the support **190** is biased.

FIG. **10** is a diagram illustrating an example of an eye feature identified from an array of light received through the light receiver **1520** when a light emitter of the AR device **1000** is the IR scanner **1520-1** or **1520-2** according to an example embodiment of the disclosure.

Referring to FIG. **10**, the IR scanner **1520-1** before the support **190** of the glasses type AR device **1000** as shown in FIG. **2** is biased may sequentially emit IR light of point lint toward a region where user's eyes are located by using the light reflector **1400**, and the light receiver **1520** may receive a third light array **100**.

In addition, after the support **190** of the AR device **1000** is biased, the IR scanner **1520-2** may sequentially emit IR light of point lint toward the region where user's eyes are located by using the light reflector **1400**, and the light receiver **1520** may receive a fourth light array **102**.

In the third light array **100** and the fourth light array **102**, positions and spaces of light signals corresponding to feature points of eyes may be different from each other. The AR device **1000** may calibrate positions of parts corresponding to the feature points of the eyes from the fourth light array **102** in consideration of a bias angle of the support **190**.

For example, the processor **1800** may identify parts **102-1** in the fourth light array **102** corresponding to the glint feature points of the eyes, based on brightness of lights in the fourth light array **102**, and identify coordinate values **102-2** on the coordinate system of the IR scanner indicating the parts **102-1** corresponding to the glint feature points of the eyes. Thereafter, the processor **1800** may calibrate the coordinate values **102-2** indicating the glint feature points of the eyes by using the bias angle calculated in FIG. **9**. For example, the processor **1800** may multiply the coordinate values **102-2** representing the glint feature points of the eyes by a compensation matrix **18** of FIG. **11** which will be described below, thereby obtaining calibrated coordinate values.

In FIG. **10**, for convenience of description, the IR scanner **1520-1** before the support **190** is biased and the IR scanner **1520-2** after the support **190** of the AR device **1000** is biased are illustrated separately in FIG. **10**, in order to distinguish the IR scanner **1520-1** before the support **190** of the AR device **1000** is biased from the IR scanner **1520-2** after the support **190** of the AR device **1000** is biased. However, the IR scanner **1520-1** before the support **190** is biased and the IR scanner **1520-2** after the support **190** of the AR device **1000** are the same IR scanners installed in the AR device **1000**.

In addition, in FIG. **10**, for convenience of description, it has been described that third light array **100** is used to identify feature points of eyes using the IR scanner **1520-1** before the support **190** is biased, and one fourth light array **102** is used to identify feature points of eyes using the IR scanner **1520-2** after the support **190** of the AR device **1000** is biased, but the disclosure is not limited thereto. A plurality of light arrays for covering user's eyes may be used to identify feature points of eyes using the IR scanner **1520-1** before the support **190** is biased, and a plurality of light arrays for covering user's eyes may be used to identify feature points of eyes using the IR scanner **1520-2** after the support **190** is biased.

FIG. **11** is a diagram illustrating examples of functions used by the AR device **1000** to calculate a center of an eyeball and calculate a gaze point **16** of a user according to an example embodiment of the disclosure.

Equation 11 represents a relationship between a coordinate value **12** of a pupil center of the eye in a coordinate system of an IR camera and a coordinate value **13** in real space representing the center of the eyeball. In an example embodiment of the disclosure, the coordinate value **12** of the coordinate system of the IR camera may have a 2D coordinate value, and the coordinate value **13** of the real space may have a 2D coordinate value or a 3D coordinate value.

For example, when the coordinate value **13** representing the center of the eyeball is multiplied by a camera rotation matrix **20** and a scale factor and a value representing a bias in an image is added, the coordinate value **12** of the pupil center of the eye may be calculated. The camera rotation matrix **20** is a matrix that converts a coordinate value in real space into a coordinate value of a camera coordinate system in consideration of a position in which a camera is provided. The coordinate value **13** representing the center of the eyeball may be converted into the coordinate value of the coordinate system of the IR camera by multiplying the coordinate value **13** representing the center of the eyeball by the camera rotation matrix **20**, and the size of the converted coordinate value may be normalized by multiplying the converted coordinate value by a scale factor. In addition, by adding a value indicating the bias in the image to the normalized size of the coordinate value, a normalized 2D coordinate value may be corrected by reflecting the bias in the image so that the coordinate value **12** of the pupil center of the eye may be calculated. In addition, the value representing the bias in the image may be used to arrange a position of the eye in an image captured by the IR camera in a reference position. For example, the value representing the bias in the image may be a value for moving a center point of the eye in the captured image to a center point of the captured image, based on a difference between the center point of the image captured by the IR camera and the center point of the eye in the captured image.

In an example embodiment of the disclosure, the camera rotation matrix **20**, the scale factor and the value representing the bias in the image may be determined in consideration of, for example, a position in which an IR LED is provided, a position in which the IR camera is provided, a capturing direction of the IR camera, an angle of view of the IR camera, the image captured by the IR camera, information related to the human eye (e.g., eyeball size, pupil size, etc.), previously captured eye images, etc., and may be previously set in the AR device **1000** during manufacturing of the AR device **1000**.

In an example embodiment of the disclosure, the AR device **1000** may input the coordinate value **12** of the pupil center of the eye into Equation 11, thereby obtaining the coordinate value **13** representing the center of the eyeball.

Equation 15 may represent a relationship between values **17** representing feature points of the eye and the gaze point **16** of the user. For example, the feature points calibrated by reflecting a bias of the support **190** may be obtained by multiplying the values **17** representing the feature points of the eye by a compensation matrix **18**. In addition, a value **19** that is output by inputting a value obtained by multiplying the values **17** representing the feature points of the eye by the compensation matrix **18** into a mapping function F may be a coordinate value **16** representing the gaze point of the user. The compensation matrix **18** may be a matrix for compensating for degree of a bias of the support **190**. The compensation matrix **18** may be determined through comparison between images captured from the IR camera in a state in which the support **190** of the AR device **1000** of FIG. **2** is not biased and images captured by the IR camera in a state in which the support **190** of the AR device **1000** is biased, and the compensation matrix **18** may be previously set in the AR device **1000** during manufacturing of the AR device **1000**. For example, the compensation matrix **18** may be determined so that the images captured from the IR camera in a state in which the support **190** is biased may be converted into the images captured by the IR camera in a state in which the support **190** is not biased, in consideration of the degree of the bias of the support **190**. However, the example in which the compensation matrix **18** is determined is not limited thereto.

For example, the compensation matrix **18** may be determined through comparison between information obtained from the images captured from the IR camera in a state in which the support **190** is not biased and information obtained from the images captured by the IR camera in a state in which the support **190** of the AR device **1000** is biased. Data obtained from an image may include, for example, a size of the image, a position of the eye in the image, a position of the pupil in the image, a position of a pattern in the image, etc., but the disclosure is not limited thereto.

Alternatively, for example, the compensation matrix **18** may be determined by comparing the images captured from the IR camera in a state in which the support **190** is not biased and the information obtained from the images captured by the IR camera in a state in which the support **190** of the AR device **1000** is biased. Alternatively, for example, the compensation matrix **18** may be determined by comparing the information obtained from the images captured from the IR camera in a state in which the support **190** is not biased and the images captured by the IR camera in a state in which the support **190** of the AR device **1000** is biased.

In addition, the mapping function F, which is a function for calculating the gaze point of the user from the feature points of the eye, may be determined such that the gaze point of the user is calculated from the reward matrix **18** and the feature points of the eye, and may be previously set in the AR device **1000** during manufacturing of the AR device **1000**.

The AR device **1000** may obtain the coordinate value **16** representing the gaze point of the user from positions of the feature points of the eye using Equation 15.

FIG. **12** is a flowchart of a method, performed by the AR device **1000** of FIGS. **2** and **3**, of detecting a user's gaze according to an example embodiment of the disclosure.

In operation S**1200**, the AR device **1000** may emit IR light toward the light reflector **1400** through the light emitter **1510** installed on the support **190** extending from the frame **110** of the AR device **1000**. The AR device **1000** may emit the IR light toward at least a partial region of the light reflector **1400** so that the IR light reflected by the light reflector **1400** may cover user's eyes.

For example, when the light receiver **1520** is an IR camera, the light emitter **1510** may be an IR LED, and the AR device **1000** may control the IR LED so that the IR light emitted from the IR LED may be reflected by the light reflector **1400** and may cover the user's eyes, in order for the IR camera to capture the user's eyes. Alternatively, for example, when the light receiver **1520** is an IR detector, the light emitter **1510** may be an IR scanner, and the AR device **1000** may control the IR scanner to scan the user's eyes by reflecting the IR light emitted from the IR scanner by using the light reflector **1400**, so that the IR detector may detect the user's eyes.

In operation S**1205**, the AR device **1000** may receive the IR light reflected by the user's eye and reflected back by the light reflector **1400**, through the light receiver **1520** installed on the support **190** extending from the frame **110** of the AR device **1000**.

For example, when the light emitter **1510** is an IR LED, the light receiver **1520** may be an IR camera, and the AR device **1000** may control the IR camera to capture the user's eyes through the light reflected by the light reflector **1400** from the user's eyes. Alternatively, for example, when the light emitter **1510** is an IR scanner, the light receiver **1520** may be an IR detector, and the AR device **1000** may control the IR detector to detect the IR light reflected by the user's eye and reflected back by the light reflector **1400**, so that the IR detector may detect the user's eyes.

In operation S**1210**, the AR device **1000** may detect previously set features related to the gaze of the user's eyes based on the received IR light. For example, the AR device **1000** may detect a position of a pupil feature point of the user's eyes and a position of a glint feature point of the eyes. The pupil feature point may be, for example, a pupil central point, and the glint feature point of the eyes may be a part having brightness greater than or equal to a certain value in a detected eye region. The position of the pupil feature point and the position of the glint feature point of the eyes may be identified, for example, by a coordinate value indicating a position in a coordinate system of the light receiver **1520**. For example, the coordinate system of the light receiver **1520** may be a coordinate system of an IR camera or a coordinate system of the IR detector, and the coordinate value in the coordinate system of the light receiver **1520** may be a 2D coordinate value.

The AR device **1000** may detect previously set features related to the gaze of the eyes by analyzing the light received by the light receiver **1520**. For example, when the light receiver **1520** is an IR camera, the AR device **1000** may identify the position of the pupil feature point and the position of the glint feature point of the eyes in an image captured by the IR camera. Alternatively, for example, when the light receiver **1520** is an IR detector, the AR device **1000** may analyze the IR light detected by the IR detector, thereby identifying the position of the pupil feature point and the position of the glint feature point of the eyes.

In addition, the AR device **1000** may analyze the light received by the light receiver **1520**, thereby obtaining a coordinate value indicating the position of the pupil feature point and a coordinate value indicating the position of the glint feature point of the eyes. For example, when the light receiver **1520** is an IR camera, the AR device **1000** may obtain the coordinate value of the pupil feature point and the coordinate value of the glint feature point of the eyes from the coordinate system of the IR camera. For example, when the light receiver **1520** is an IR camera, the AR device **1000** may identify the position of the pupil central point in an image captured by the IR camera. For example, the position of the pupil central point may have a coordinate value in the coordinate system of the IR camera.

For example, the AR device **1000** may identify a position of the brightest point in the image captured by the IR camera, in order to identify the glint feature point of the eyes. The AR device **1000** may identify the brightness of the IR light received through an image sensor of the IR camera including a plurality of photodiodes, and may identify at least one pixel corresponding to bright IR light equal to or greater than a certain reference among pixels of the image captured by the IR camera, thereby identifying the position of the glint feature point of the eyes. For example, the AR device **1000** may identify the pixel corresponding to the brightest IR light among the pixels of the image captured by the IR camera, thereby identifying the position of the glint feature point of the eyes. For example, the position of the glint feature point of the eyes may have a coordinate value in the coordinate system of the IR camera.

Alternatively, for example, when the light receiver **1520** is an IR detector, the AR device **1000** may calculate the coordinate value of the pupil feature point and the coordinate value of the glint feature point of the eyes in the coordinate system of the IR detector.

When the light emitter **1510** is an IR scanner, the AR device **1000** may control the IR scanner to sequentially irradiate a point light source ora line light source to cover a region where the user's eyes are located, and sequentially receive the light reflected from the user's eyes through the IR detector in order to scan the region where the user's eyes are located. Also, the AR device **1000** may analyze an array of light sequentially received through the IR detector, thereby identifying the pupil feature point and the glint feature point of the eyes. For example, the AR device **1000** may identify light having a brightness equal to or greater than a certain value in the received light array, thereby identifying a coordinate of the glint feature point of the eyes. For example, the position of the glint feature point of the eyes may have a coordinate value in the coordinate system of the IR detector.

In operation S**1215**, the AR device **1000** may detect a pattern of the light reflector **1400** based on the received IR light. The light reflector **1400** may be coated on one surface of the waveguide **170** of the AR device **1000** to have a certain pattern. The AR device **1000** may receive the IR light reflected by the user's eyes and reflected by the light reflector **1400** through the light receiver **1520**, and identify a shape of the pattern based on the received IR light. The pattern formed on the light reflector **1400** may include, for example, a dot pattern, a line pattern, a grid pattern, a 2D marker, etc., but the disclosure is not limited thereto. When the temple **191** is biased with respect to the frame **110**, the pattern identified by the pattern detection code **1740** may have a deformed shape.

For example, when the light receiver **1520** is an IR camera, the IR camera may capture the user's eyes based on the IR light reflected by the light reflector **1400**, and the AR device **1000** may identify a pattern within an image obtained by capturing the user's eyes from the image. For example, when the light receiver **1520** is an IR detector, the IR detector may sequentially receive IR lights reflected by the light reflector **1400**, and the AR device **1000** may identify a part related to the pattern of the light reflector **1400** in an array of the sequentially received IR lights.

In operation S**1220**, the AR device **1000** may determine a degree to which the support **190** of the AR device **1000** is biased with respect to the frame **110**. When the IR light is received after the support **190** is biased with respect to the frame **110**, the AR device **1000** may identify a pattern having a deformed shape from the received IR light. Also, for example, as in FIGS. **8**A and **8**B, the AR device **1000** may compare the pattern having the deformed shape with a non-deformed pattern, thereby estimating the degree of bias of the support **190**. For example, the degree of bias of the temple **191** may be expressed as a bias angle indicating a difference between a default angle of the temple **191** with respect to the frame **110** and an angle of the biased temple **191** with respect to the frame **110**, but the disclosure is not limited thereto. Also, for example, the degree of bias of the nose support **192** may be expressed as a bias angle indicating a difference between a default angle of the nose support **192** with respect to the frame **110** and an angle of the biased nose support **192** with respect to the frame **110**, but the disclosure is not limited thereto.

Also, for example, when the light receiver **1520** is an IR detector, the coordinate value of the pupil feature point and the coordinate value of the glint feature point of the eyes may be values calibrated by reflecting the degree of bias of the support **190** of the AR device **1000**. When the light receiver **1520** is an IR detector, for example, when the coordinate values **102-2** corresponding to the IR lights **102-1** corresponding to the feature points of the eyes in the light array **102** of FIG. **10** are calculated, the degree of bias of the support **190** may be reflected. For example, the AR device **1000** may calibrate positions of the lights **102-1** corresponding to the feature points of the eyes in the light array **102** among the lights in the light array **102** received by the IR detector, by reflecting the degree of bias of the support **190**. The AR device **1000** may directly calculate the coordinate values **102** corresponding to the feature points of the eyes in the coordinate system of the IR detector, based on the calibrated positions of the lights **102-1** corresponding to the feature points of the eyes in the light array **102**. In this case, the AR device **1000** may calculate the coordinate value of the pupil feature point and the coordinate value of the glint feature point of the eyes calibrated by reflecting the degree of bias of the temple **191** of the AR device **1000** and/or the degree of bias of the nose support **192**. The calibrated coordinate values may be input to a mapping function.

In operation S**1225**, the AR device **1000** may identify the pupil position of the user's eyes based on the IR light reflected from the light reflector **1400**. For example, when the light receiver **1520** is an IR camera, the AR device **1000** may identify the pupil position of the user's eyes within an image captured by the IR camera from the image. Alternatively, for example, when the light receiver **1520** is an IR detector, the AR device **1000** may analyze the IR light sequentially obtained by the IR detector, thereby calculating the pupil position of the user's eyes. The AR device **1000** may identify the pupil central point of the user's eyes, thereby identifying the pupil position of the user's eyes.

In operation S**1230**, the AR device **1000** may obtain a gaze direction of the user. The AR device **1000** may calculate a position of the center of the user's eyes. The center of the user's eyes may be the center of user's eyeballs. The AR device **1000** may calculate the position of the center of the user's eyes, based on the pupil position of the user's eyes and the degree of bias of the support **190**. For example, the processor **1800** may calculate the position of the center of the user's eyes so that a value calculated based on a matrix for calibrating the degree of bias of the support **190**, a value indicating the position of the center of the user's eyes and a bias of axis of the image obtained by capturing the user's eyes can be a value of the pupil position of the user's eyes obtained by the pupil position detection code **1760**. For example, the center of the eye may be the center of the eyeball, and the position of the center of the user's eyes may have a 3D coordinate value in a coordinate system of a real space.

The AR device **1000** may calculate a position of the gaze point of the user. In order to calculate the position of the gaze point of the user, the AR device **1000** may previously generate a mapping function for calculating the position of the gaze point from features of the user's eyes. The mapping function is a function for calculating the position of the gaze point of the user in consideration of features of the user's eyes and bias information of the support **190**, and may be generated during a calibration process of the calibration code **1780**. For example, the position of the gaze point may have a 3D coordinate value in the coordinate system in the real space, but the disclosure is not limited thereto. For example, the position of the gaze point may have a coordinate value in the coordinate system of the waveguide **170**, but is not limited thereto.

The AR device **1000** may calibrate the features related to the user's gaze based on the degree of bias obtained from the bias determination code **1750**. Also, the AR device **1000** may apply the features related to the user's gaze calibrated based on the degree of bias to the mapping function, thereby calculating the position of the gaze point of the user. Also, a gaze direction of the user may be determined based on the position of the central point of the user's eyes and the gaze point of the user calculated by the gaze determination code **1770**.

Meanwhile, the AR device **1000** may calibrate the mapping function based on the bias angle of the support **190**. The AR device **1000** may calibrate the mapping function for obtaining the gaze point of the user based on a default bias angle and features of eyes.

For example, when the light receiver **1520** is an IR camera, the AR device **1000** may display a target point for calibration through the waveguide**170**, and capture the user's eyes looking at the target point by using the IR camera. In addition, the processor **1800** may identify and analyze a pattern within the image obtained by capturing the user's eyes, thereby obtaining a bias angle of the support **190**. In addition, the AR device **1000** may detect positions of the feature points related to the user's eyes from the image obtained by capturing the user's eyes, and input the positions of the feature points of the user's eyes and the bias angle of the support **190** into the mapping function. The AR device **1000** may calibrate the mapping function so that a position value of the target point may be output from the mapping function to which the positions of the feature points of the user's eyes and the bias angle of the support **190** are input.

For example, when the light receiver **1520** is an IR detector, the AR device **1000** may display the target point for calibration on the waveguide**170** and control the IR scanner to emit IR light for scanning the user's eyes looking at the target point. In addition, the AR device **1000** may receive and analyze IR light reflected from the user's eyes through the IR detector, identify a pattern of the light reflector **1400**, and estimate the bias angle of the temple **191**. The AR device **1000** analyze the IR light based on the bias angle of the temple **191**, thereby identifying positions of the calibrated feature points of eyes. For example, when the AR device **1000** estimates the positions of the feature points of eyes by using the IR scanner and the IR detector, because results of estimating the positions of the feature points of eyes are affected by an operating angle of the IR scanner, the positions of the feature points of eyes may be calibrated based on the bias of the support **190** by calculating a value obtained by subtracting the bias angle of the support **190** from the operating angle of the IR scanner.

In addition, the AR device **1000** may input the positions of the calibrated feature points of the eyes into the mapping function, and calibrate the mapping function so that the position value of the target point may be output from the mapping function in which the calibrated positions of the feature points of the eyes are input.

An example embodiment of the disclosure may be implemented as a recording medium including computer-readable instructions such as a computer-executable program module. According to an example embodiment, the computer-readable instructions may be computer codes, which are implemented as one or more computer-executable program modules. A computer-readable medium may be any available medium which is accessible by a computer, and may include a volatile or non-volatile medium and a removable or non-removable medium. Also, the computer-readable medium may include computer storage medium and communication medium. The computer storage media include both volatile and non-volatile, removable and non-removable media implemented in any method or technique for storing information such as computer readable instructions, data structures, program codes, program modules or other data. The communication medium may typically include computer-readable instructions, data structures, or other data of a modulated data signal such as program modules.

A computer-readable storage medium may be provided in a form of a non-transitory storage medium. Here, the term 'non-transitory storage medium' refers to a tangible device and does not include a signal (e.g., an electromagnetic wave), and does not distinguish between a case where data is stored in a storage medium semi-permanently and a case where data is stored temporarily. For example, the non-transitory storage medium may include a buffer in which data is temporarily stored.

According to an example embodiment of the disclosure, the method according to various embodiments of the disclosure disclosed herein may be included in a computer program product and provided. The computer program product may be traded between a seller and a purchaser as a commodity. The computer program product may be distributed in a form of a machine-readable storage medium (e.g., compact disk read only memory (CD-ROM)), or may be distributed online (e.g., downloaded or uploaded) through an application store (e.g., PlayStore™) or directly between two user devices (e.g., smart phones). In the case of online distribution, at least a portion of the computer program product (e.g., a downloadable app) may be temporarily stored in a machine-readable storage medium such as a manufacturer's server, an application store's server, or a memory of a relay server.

In addition, in the specification, the term "unit" may be a hardware component such as a processor or a circuit, and/or a software component executed by a hardware component such as a processor.

Also, in the specification, the expression "include at least one of a, b or c" means "include only a", "include only b", "include only c", "include a and b", "include b and c", "include a and c", or "include a, b, and c".

The above-described description of the disclosure is provided only for illustrative purposes, and those of skill in the art will understand that the disclosure may be easily modified into other detailed configurations without modifying technical aspects and essential features of the disclosure. Therefore, it should be understood that the above-described embodiments are exemplary in all respects and are not limited. For example, the elements described as single entities may be distributed in implementation, and similarly, the elements described as distributed may be combined in implementation.

The scope of the disclosure is not defined by the detailed description of the disclosure but by the following claims, and all modifications or alternatives derived from the scope and spirit of the claims and equivalents thereof fall within the scope of the disclosure.

The invention claimed is:

1. An augmented reality (AR) device comprising:
- a waveguide;
- a light reflector on which a pattern is formed;
- a support configured to secure the AR device to a face of a user, the support comprising a temple extending from a frame around the waveguide to be positioned on an ear of the user;
- a light emitter and a light receiver installed on the support; and
- at least one processor configured to:
  - control the light emitter to emit first light toward the light reflector on which the pattern is formed,
  - receive, through the light receiver, second light reflected by an eye of the user and reflected by the light reflector,
  - identify a reflected pattern corresponding to the pattern formed on the light reflector from the second light,
  - identify a degree of bias of the temple with respect to the frame by comparing the identified reflected pattern with a pattern identified from a light received by the light receiver before the temple is biased,
  - obtain gaze information of a user of the AR device based on the identified reflected pattern, and
  - perform calibration on the gaze information based on the degree of bias of the temple with respect to the frame,
- wherein the first light emitted toward the light reflector is reflected by the light reflector and directed toward the eye of the user, and
- wherein the second light received by the light receiver comprises light from the first light directed toward the eye of the user being reflected by the eye of the user, reflected back by the light reflector, and directed toward the light receiver.

2. The AR device of claim **1**, wherein the light reflector is formed on the waveguide.

3. The AR device of claim **1**, wherein the at least one processor is further configured to:
- generate a mapping function for calculating a position of a gaze point of the user based on the degree of bias of the support with respect to the frame, and

based on the mapping function and the degree of bias of the support with respect to the frame, obtain the gaze information of the user.

4. The AR device of claim **3**, wherein the at least one processor is further configured to, based on the second light received by the light receiver, obtain a position of one or more feature points corresponding to the eye of the user.

5. The AR device of claim **4**, wherein the at least one processor is configured to input the position of the one or more feature points corresponding to the eye of the user and the degree of bias of the support with respect to the frame into the mapping function and calculate the position of the gaze point of the user.

6. The AR device of claim **3**, wherein the at least one processor is further configured to:

display a target point at a specific position on the waveguide in order to calibrate the mapping function,

receive third light reflected by the eye of the user looking at the displayed target point through the light receiver, and

calibrate the mapping function based on the third light reflected by the eye of the user looking at the displayed target point.

7. The AR device of claim **6**, wherein, for calibration of the mapping function, the at least one processor is further configured to:

based on the third light reflected by the eye of the user looking at the displayed target point, identify the reflected pattern of the light reflector,

based on the identified reflected pattern, identify the degree of bias of the support, and

based on the third light reflected by the eye of the user looking at the displayed target point, obtain a position of one or more feature points corresponding to the eye of the user looking at the displayed target point.

8. The AR device of claim **7**, wherein the at least one processor is further configured to:

input the degree of bias of the temple and the position of the one or more feature points into the mapping function, and

calibrate the mapping function so that a position value of the target point is output from the mapping function.

9. The AR device of claim **1**, wherein the light reflector is coated on the waveguide.

10. The AR device of claim **1**, wherein the light reflector comprises at least one of silver, gold, or copper.

11. The AR device of claim **1**, wherein the pattern formed on the light reflector comprises one of a dot pattern, a line pattern, a grid pattern, or a 2D marker.

12. A method, performed by an augmented reality (AR) device, of detecting a gaze of a user, the method comprising:

emitting, by a light emitter installed in a support of the AR device, first light toward a light reflector on which a pattern is formed, the first light emitted by the light emitter being directed toward an eye of a user wearing the AR device, the support comprising a temple extending from a frame around a waveguide to be positioned on an ear of the user;

receiving, by a light receiver installed on the support configured to secure the AR device to a face of the user, second light reflected by the eye of the user and reflected by the light reflector;

identifying a reflected pattern corresponding to the pattern formed on the light reflector from the second light;

identifying a degree of bias of the temple with respect to the frame by comparing the identified reflected pattern with a pattern identified from a light received by the light receiver before the temple is biased;

obtaining gaze information of the user based on the identified pattern; and

calibrating the gaze information based on the degree of bias of the temple with respect to the frame,

wherein the second light received by the light receiver comprises light from the first light directed toward the eye of the user being reflected by the eye of the user, reflected back by the light reflector, and directed toward the light receiver.

13. The method of claim **12**, further comprising generating a mapping function for calculating a position of a gaze point of the user based on the degree of bias of the support with respect to the frame,

wherein the calibrating of the gaze information comprises calculating the position of the gaze point of the user based on the mapping function and the degree of bias of the support with respect to the frame.

14. The method of claim **13**, further comprising, based on the second light received by the light receiver, obtaining a position of one or more feature points corresponding to the eye of the user,

wherein the calibrating of the gaze information comprises inputting the position of the one or more feature points corresponding to the eye of the user and the degree of bias of the support with respect to the frame into the mapping function and calculating the position of the gaze point of the user.

15. A non-transitory computer-readable recording medium having recorded thereon a program for executing the method of claim **11** on a computer.

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